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Launch 2 Learn

Introductory Rocket Workshop Handbook

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2. Workshop Supply List

2a. Virtual Workshop

If you are attending a virtual workshop, you have received a supply box with all of the items required to build your rocket. **IT IS YOUR RESPONSIBILITY TO ATTEND THE *LAUNCH 2 LEARN KIT REVEAL MEETING* TO VERIFY YOUR SUPPLY KIT IS COMPLETE**. WSGC will issue missing components based upon the reveal meeting. A checklist is provided on the next page.

In addition to the supplies we provided, you will also need to secure a quiet workspace with ample table space, a computer with a camera and microphone, and reliable internet. If you have concerns about your ability to participate in the workshop, please contact us as soon as possible at spacegrant@carthage.edu.

If you are missing any items, email Rob Cannon as soon as possible at rcannon@carthage.edu with the Subject Line: "Launch 2 Learn Workshop – Missing Supplies."

2b. Workshop Supply Check-List

Item Description:	Quantity:	Received:
Launch 2 Learn Handbook*	1	
Alcohol Wipes	15	
Awl, drill, and bits	1	
Craft Sticks, Wooden	15	
Cups, Plastic	15	
18” Wooden Dowel	1	
Epoxy	1	
Nitrile Gloves	6 pr.	
Masking Tape	1	
Paper Plates	2	
Paper Towels	bundle	
Pencil	1	
Ruler	1	
Sandpaper	1	
Screwdriver set	1	
Table Covering	1	
Utility Knife	1	

Table 2-1 Workshop Supply Checklist

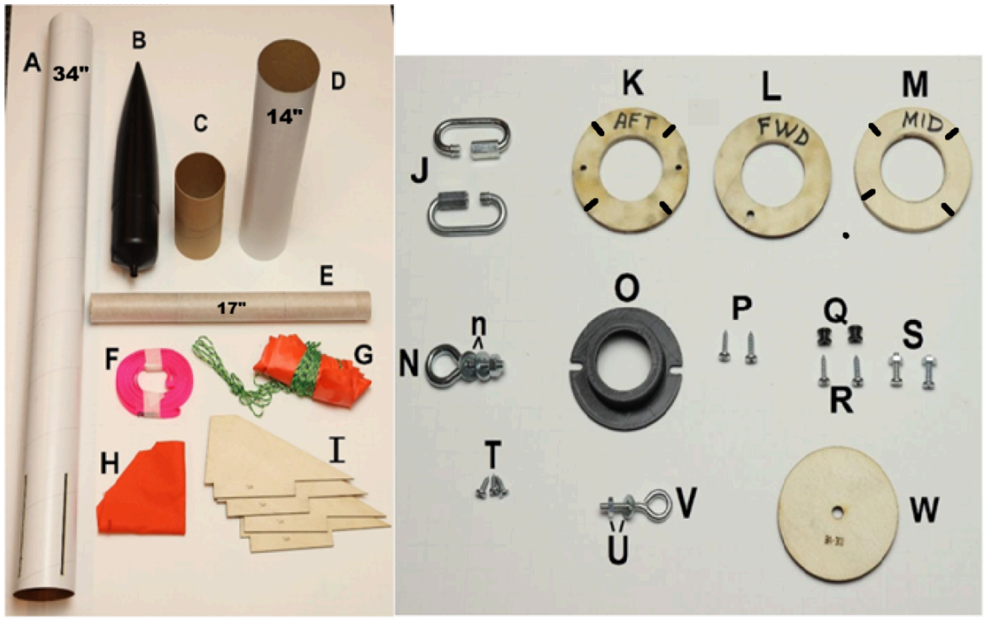


Figure 2-1: Parts included with the LOC Precision Launch 2 Learn Rocket Kit (labels correspond to parts list in Table 1)

Workshop Supply Check-List (Continued)

Item Description:	Quantity:	Received:
LOC Precision – Launch 2 Learn Rocket Kit*:	1	
A - Booster Airframe	1	
B - Nose Cone	1	
C - Coupler	1	
D - Payload Section	1	
E - 38mm Motor Mount Tube	1	
F - Recovery Strap	1	
G - Parachute	1	
H - Parachute Protector	1	
I - Fin	4	
J - Quick Link	2	
Centering Ring Bag:		
K - Centering Ring with 2 Small Holes	1	
L - Centering Ring with 1 Small Hole	1	
M - Centering Ring with 0 Small Holes	1	
N - Big Eyebolt	1	
n - 2 Washers, 1 Nut	2	
O - Motor Retention Cap	1	
P - #8 Wood Screws	2	
Rail Button Bag:		
Q - Rail Button	2	
R - Wood Screw	2	
S - Machine Screw w/nut	2	
Bulk Plate Bag:		
U - Nut w/ washer	2	
V - Small Eyebolt	1	
W - Bulk Plate	1	

Table 2 Workshop Supply Checklist Launch 2 Learn Rocket Kit

3. Workshop Objectives

The *Launch 2 Learn Introductory Rocket Workshop* has multiple objectives:

- Introduce teams to the design, build, fly stages of high-powered rocket
- Build a Level 1 high-powered single deploy kit rocket
- Develop basic understanding of rocket flight simulation
- Understand High Power Rocketry Level 1 certification

4. High-Powered Rocketry Overview

4a. Rocketry History and Organizations

Amateur high-powered rocketry (HPR) began in the 1950's following the launch of Sputnik and coinciding with the Space Race.

Two national organizations support high-powered rocketry in the United States:

Tripoli Rocketry Association (TRA) http://www.tripoli.org

National Association of Rocketry (NAR) https://www.nar.org

In Canada: Canadian Association of Rocketry (CAR) https://www.canadianrocketry.org

To launch an HPR safely and legally, you must belong to one of these clubs, and follow the necessary procedures as determined by the club.

4b. Launch Profile Overview

The typical launch profiles for high-powered rockets are:

1. Launch
2. Powered Ascent
3. Coasting Flight
4. Ejection Charge/Drogue Deploy
5. Slow Descent
6. Ejection Charge/Main Deploy
7. Recovery

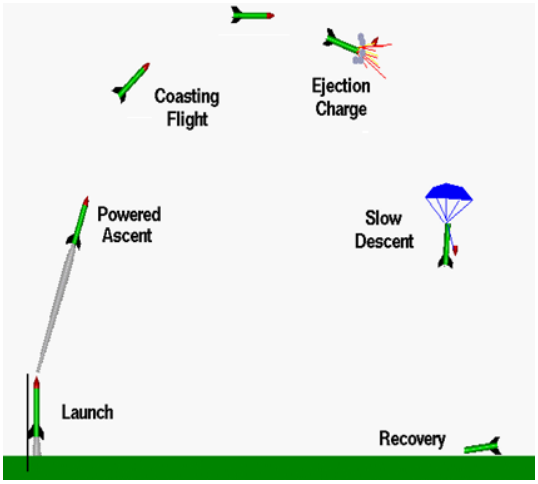


Figure 4-1 Single Deploy Flight Profile

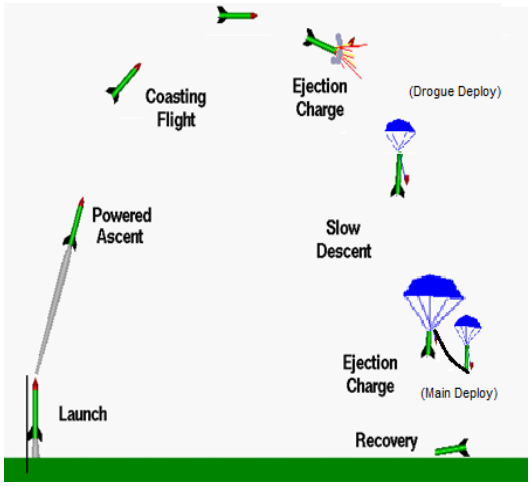


Figure 4-2 Dual Deploy Flight Profile

4c. Rocketry Resources

There are many resources available online for the new amateur rocket enthusiast. Many manufacturers make rocket kits that you can build for HPR. Some of these include:

- Aerotech
- LOC Precision
- Madcow Rocketry
- Public Missiles Limited (PML)

Many vendors distribute HPR kits along with other necessary rocketry supplies (think one-stop shopping):

- Apogee Rockets
- Giant Leap Rocketry
- Wildman Rocketry

The Tripoli Rocketry Association website contains links to many rocket vendors at: http://www.tripoli.org/Vendors.

Kits come in a range of materials for a range of skill levels and budgets. Paper, phenolic, blue tube, fiberglass, and carbon fiber are all popular choices. Recovery hardware, electronics, and motors can be found at the vendors listed on the Tripoli website also. At the end of this workshop, we will recap the procedures of how to design, build, and fly an HPR.

4d. High-Powered Rocketry Overview Notes

5. Structures

The main component of a high-powered rocket build is the airframe structure itself. Other systems include recovery and propulsion. The size and internal structure of an amateur high-powered rocket is different from actual rockets that attain suborbital and orbital trajectories. This section discusses the main structural components of high-powered rockets.

5a. Parts and Subassemblies *(Build Reference 13.a)*

All high-powered rockets have a few key elements that are easily recognized from the exterior: fins, a body tube, and a nose cone. There are also other external elements such as rail buttons and motor retainers. There are other features in the interior of the rocket; couplers, bulkheads, centering rings, motor mount tubes, recovery hardware etc.

A typical high-powered rocket will consist of an aft section (the booster) and a forward section (the sustainer) joined by a coupler.

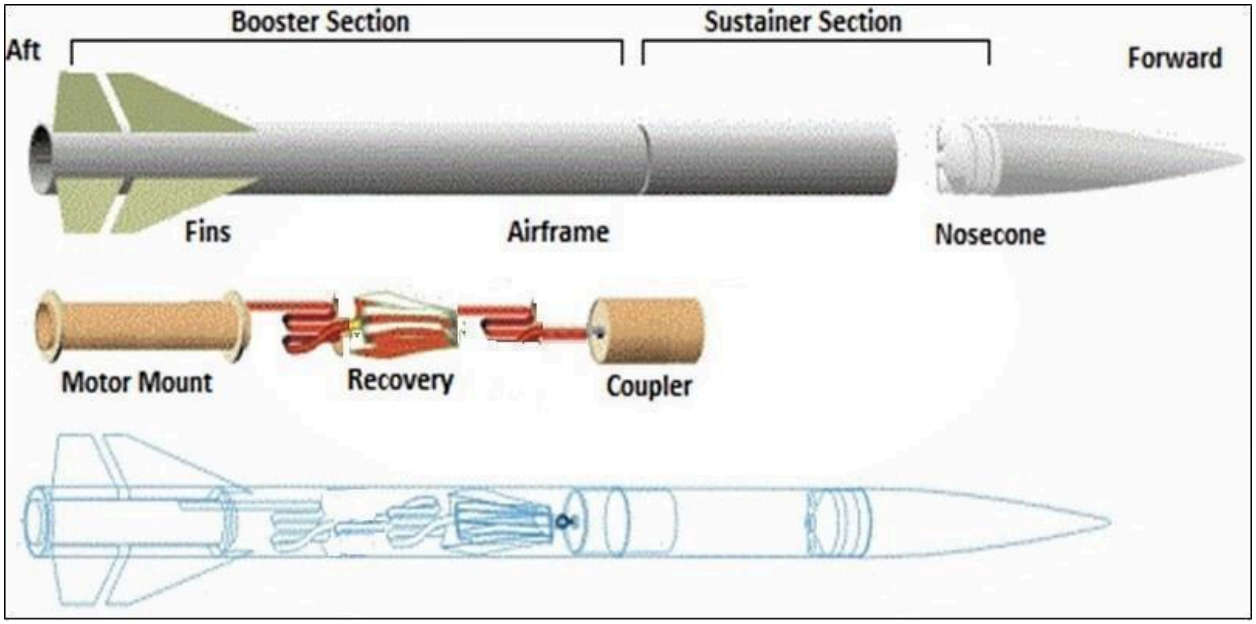


Figure 5-1: High-Powered Rocket with Coupler

For more details about high-powered rocket structures, please see the NASA High-Powered Video Series Counterpart document, pages 3 – 16.

https://www.youtube.com/playlist?list=PLiuUQ9asub3Rikic9KVsF2RjVo7rB3dwU

https://www.nasa.gov/sites/default/files/atoms/files/sl_video_instruction_book.pdf

5b. Nose Cone *(Build references 13.b.vi)*

The nose cone subassembly consists of the nose cone and any interior features that would attach it to the rest of the rocket (such as bulkheads, eyebolts, etc.). In certain cases, the nose cone can be part of the payload/experiment that the rocket vehicle is carrying or can contain the recovery or tracking electronics.

Nose cones are usually made of molded injection plastic, fiberglass, or carbon fiber.

5c. Airframe *(Build reference 13.a.iv)*

Rocket airframes are generally smooth thin walled cylinders with a high length to diameter ratio and encompass the rocket's propulsion system, recovery system, electronics, and payload.

Generally, rockets have a nose cone fitted to the forward end of the airframe and a set of fins mounted towards the aft end. On a *minimum diameter* rocket, the airframe also serves as the motor mount tube. This design eliminates the need for centering rings and a motor mount tube, but presents new challenges because the fins must be surface mounted.

High-powered rocket airframes are typically made of non-metallic, high strength to weight ratio composite materials like carbon fiber, fiberglass, phenolic and PVC (NAR rules). Paper and cardboard will not handle the loads of high-powered rocket motors unless they are sufficiently reinforced with composite materials.

5d. Coupler *(Build references 13.a.ii; 13.b.iii, iv)*

A coupler can be used to join multiple segments of the airframe. A coupler is a tube that's outer diameter (OD) is equal to the inner diameter (ID) of the airframe, allowing for a snug fit inside. When joining airframes with a coupler, the coupler should extend at least one airframe diameter into each joined segment. Hence, if you were to join two 4-inch diameter airframes, you would need a coupler at least 8 inches long.

Figure 5-2: Avionics Bay / Airframe Coupler Tube Assembly

Couplers, because of their required lengths to join airframe sections, make a good place to store electronics such as altimeters, batteries, switches, and so on. The coupler section makes an ideal avionics bay. The *avionics bay* in Figure 5-2 is built from a 12-inch long *coupler* with a 4-inch *collar* in the center. The collar is a segment of airframe that is epoxied to the middle of the coupler. The collar allows direct access to the switches that power on the altimeters after the rocket is fully assembled. This coupler joins two airframe segments that contain the recovery system by clicking into place with plastic rivets or screws.

5e. Centering Rings / Bulkplates *(Build references 13.a.i, ii, iii; 13.b.ii, iii, x)*

Centering rings are used to align small tubes concentrically, like motor mount tubes, inside of an airframe. Centering rings can be made of many different materials and take many shapes depending on the number of inside tubes that are to be aligned.

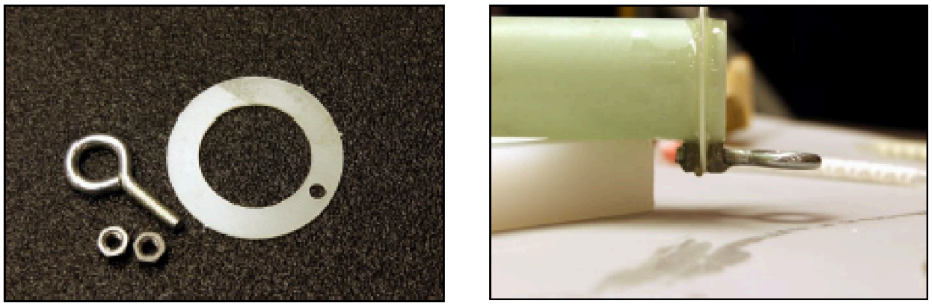


Figure 5-3: Centering Ring and Recovery Hardware

Bulkplates are circular disks that fit inside of tubes to separate volumes. Typically, centering rings and bulkplates are either made out of plywood (i.e., birch), or from composite materials i.e.,. fiberglass) and serve as the mounting structure for other systems' components such as recovery hard mounts, payloads, and electronics. Typically, there exists one centering ring above the fins and one below. Centering rings act as the primary load path, to transfer the motors thrust to the airframe.

5f. Motor Mount Tube *(Build references 13.a.iii; 13.b.ii, v)*

A motor mount tube is any tube inside of a rocket that is intended to fit a rocket motor. Motor mount tube diameters are typically called out in millimeters and refer to the motor size they are designed to hold. A typical high-powered rocket may have a 38mm, 54mm, 75mm, or 98mm motor mount tube.

Figure 5-4: Motor Mount Tube Assembly (Left); Motor and Motor Casing (Right)

Recall that on a *minimum diameter* rocket, the airframe is the motor mount tube. Generally, on smaller rockets, the motor mount tube encompasses the entire rocket motor case. Larger rocket motor mount tubes may only encompass half of the motor mount tube with the rest freely suspended inside the rocket airframe. That is because a full-length motor mount tube can be heavy, adding unnecessary weight.

5g. Motor Retention *(Build reference 13.a.i; 13.b.x)*

A motor retainer is a device that positively retains the rocket motor case inside of the rocket prohibiting it from falling out during flight. There are many commercial motor retainers available for every size motor. It is also very common for rocket builders to fashion their own motor retainer from common screws and washers from a hardware store.

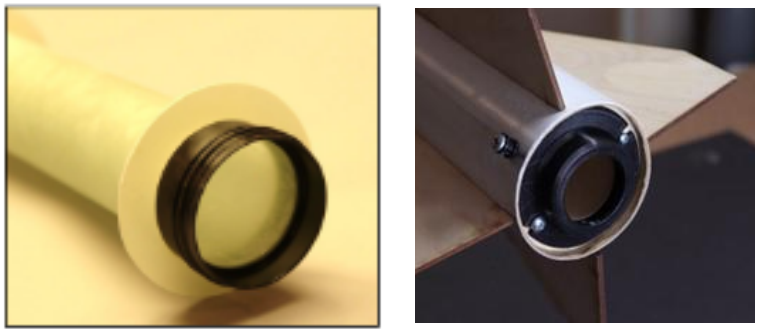


Figure 5-5: Motor Mount Tube Assembly (Left); Motor Retention cap assembly (Right)

5h. Rail Buttons (*Build references 13.b, vii, xi)*

Most high-powered rockets launch from a rail rather than a rod and require a rail button to attach the rocket to the launch rail. Rail buttons are not to be confused with launch lugs (small cylinders attached to the sides of most low-powered rockets) which attach the rocket to a launch rod. The most commonly used launch rails are 6’ tall 1010 rails (1”x1”x6’) and 10’ tall 1515 rails (1.5”x1.5”x10’). A rail can be seen in Figure 5-6.

Figure 5-6: Launch Rail

A rail button is an ‘H’ shaped component usually made of a hard, smooth plastic, that mounts to the airframe of a rocket and slides freely inside of a channel along an extruded aluminum launch rail. This system constrains the rocket’s movement until sufficient velocity is achieved that the fins become effective for flight stability.

Rail buttons should not be an afterthought of the design process. Their location should be documented as part of the vehicle's design requirement. The rocket should have one rail button near the bottom of the booster, and another near the center of gravity. It is beneficial if both rail buttons are located on the same airframe section.

Figure 5-7: Rail Button

5i. Fins *(Build references 13.a.v; 13.b.ix)*

Fins are flat fixed stabilizing structures extending from the body of a rocket that give stability in flight. The effectiveness of fins depends mainly on the size, shape, and surface finish. Fins can be many different shapes. Four common shapes are tapered, simple delta, cropped delta, and elliptic.

Fins have several features used to parametrically characterize its design; tip chord (ct), root chord (cr), span (b), leading edge (LE), trailing edge (TE), aspect ratio (AR), and tang (referenced as “tab” within RockSim). The aspect ratio is just b/ct. Most fins are tapered or delta type due to ease of manufacturing. Most rocket fins are designed with low aspect ratios (AR < 4) and taper ratio (ct ⁄ cr) between 0.2 and 0.4 is ideal for minimum induced drag (δ). The tang or tab is the part of the fin that extends inside of the airframe.

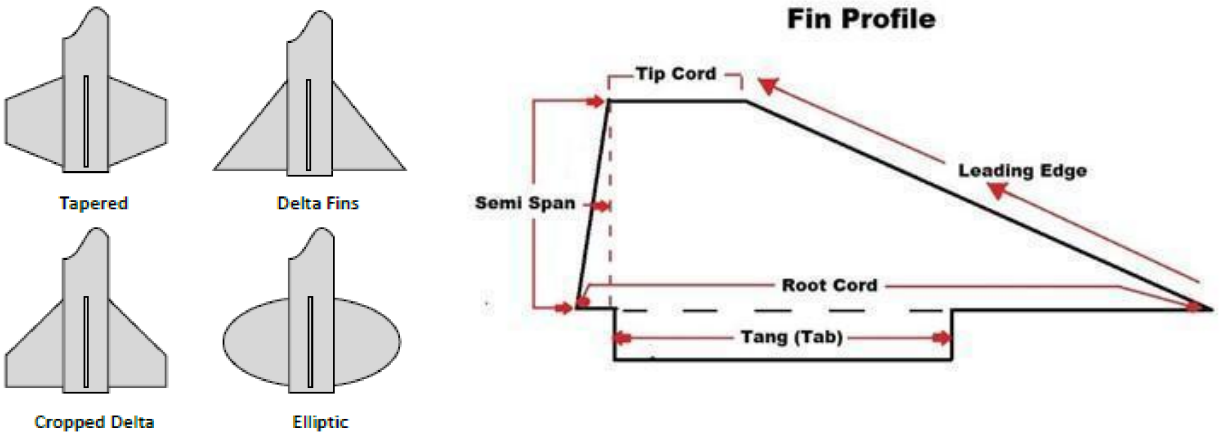


Figure 5-8: Fin Types (left) Fin components (right)

Fins are attached to a rocket using one of two methods – surface mounted (minimum diameter rockets) and through the wall (TTW). Some rockets employ removable fins. Removable fins are difficult to successfully design but you would gain multi-mission capability or just the simple ability to replace a damaged component.

Typically, fins are constructed of G10 fiberglass, carbon fiber, plastics, and/or plywood sheets cut to size. High-powered rocket fins should not be made from Plexiglas or any other non-rigid material. More complex fins can be composites of two or more of these materials.

5j. Structures Notes

6. Propulsion Systems

Rocket propulsion is a complex subject matter. Traditional propulsion systems rely on thermodynamic expansion of a supersonic gas to produce thrust. Non-traditional propulsion systems still rely on Newton, but get creative in how they produce thrust.

This workshop will only cover the basics of propulsion as they pertain to high-powered rocketry. Any aerospace propulsion textbook can be consulted for a more detailed explanation.

6a. HPR Definition and Classifications

Rockets use a propulsive device called a rocket motor that generates thrust by exhausting hot gases at high velocities. The momentum of the hot exhaust gases produces a net force in the opposite direction causing the rocket to move upwards. This happens because the rocket obeys *Newton's 3rd Law of Motion*, which states that for every action there is an equal and opposite reaction.

U.S. rocketry safety codes are published by the National Fire Protection Association (NFPA). In Canada, Transport Canada's high-power launch requirements draw on Canadian regulations, NFPA standards, and rocketry association safety codes.

A rocket exceeds the definition of a model rocket under NFPA 1122 and becomes a high-powered rocket under NFPA 1127 if it:

- Uses a motor with more than 160 Newton-seconds of total impulse (and 'H' motor or larger) or motors that all together exceed 320 Newton-seconds
- Uses a motor with more than 80 Newtons average thrust (see rocket motor coding)
- Exceeds 62.5 grams of propellant
- Weighs more than 1,500 grams including motor(s)
- Includes any airframe parts of ductile metal

HPR motors approved for sale in the United States are stamped with a two-part code that gives some basic information about the motor's power and behavior: A letter specifying the total impulse ("H"), and a number specifying the average thrust ("225"). *Average thrust* is a measure of how slowly or quickly the motor delivers its total energy and is measured in Newtons. *Total impulse* is a measure of the overall total energy contained in a motor and is measured in Newton-seconds.

Impulse Class		Category
H	160.01Ns to 320.01Ns	Level 1
I	320.01Ns to 640.00Ns	
J	640.01Ns to 1280.00Ns	Level 2
K	1280.01Ns to 2560.00Ns	
L	2560.01Ns to 5120.00Ns	
M	5120.01Ns to 10240.00Ns	Level 3
N	10240.01Ns to 20480.00Ns	
O	20480.00Ns to 40960.00Ns	

Table 6-1: HPR Motor Impulse Classes and Categories

HPR motors cannot be purchased over the counter. Members must be certified by either the National Association of Rocketry (NAR) or the Tripoli Rocketry Association (TRA) to the appropriate level to purchase motors from vendors. As part of this workshop, you may be certified NAR Level 1 pending a successful flight. You can then

purchase ‘H’ and ‘I’ motors, and one ‘J, K, or L’ on the day you intend to certify Level 2. To certify Level 2, you must pass a written test as well as conduct a successful flight.

In Canada, the Canadian Association of Rocketry (CAR) offers Levels 1–4. CAR Level 1 permits H motors and Level 2 permits I motors, splitting NAR/TRA Level 1’s H/I range. Certification motors must be certified by CAR, NAR, or TRA and accepted by Natural Resources Canada (NRCan), or covered by an NRCan general importation permit. Check authorization for the specific motor before purchase or import.

6b. Commercial Motors

For WSGC rocket teams, you are required to use a certified commercially manufactured motor. The certifying agencies are the National Association of Rocketry (NAR) and Tripoli Rocketry Association (TRA). Some of the more popular solid rocket motor manufacturers are Cesaroni Technology Inc. (CTI) and Aerotech (AT). Each manufacturer produces a wide variety of commercially certified rocket motors in all different impulse and thrust ranges.

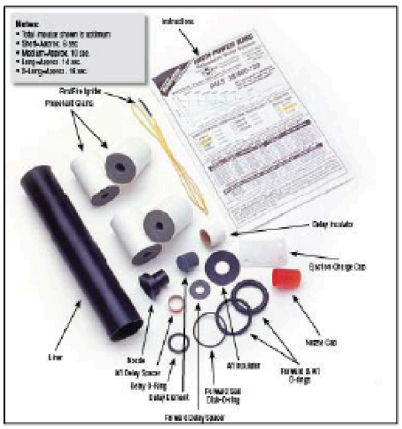


Figure 6-1: HPR Motor Components

Reloadable solid rocket motor systems are composed of a reusable motor case and a reload kit. All HPR motor cases come standard in 38mm, 54mm, 75mm, and 98mm diameters.

Most cases have three standard parts: the case, a forward closure, and an aft closure. A forward closure and a removable aft closure may be built in where the reload is loaded from on some motor cases. The closures are usually interchangeable between cases of the same diameter but in different lengths.

Figure 6-2: Cesaroni HPR Reloadable Motors

Solid rocket motor reload kits include the assembly instructions, the propellant, the nozzle, and all other one-time use hardware for the flight. A typical reload kit will include: a phenolic or paper liner that acts as thermal protection for the motor case; a series of O-rings that will seal the reload inside of the motor case; a set of phenolic disks that act as thermal protection for the aft and forward closures; a delay grain for the motor ejection charge or a tracking smoke element if the motor does not use an ejection charge; and an igniter. You will also need O-ring grease for the assembly of the reload. It is very important not to deviate from the manufacturer’s direction for motor assembly.

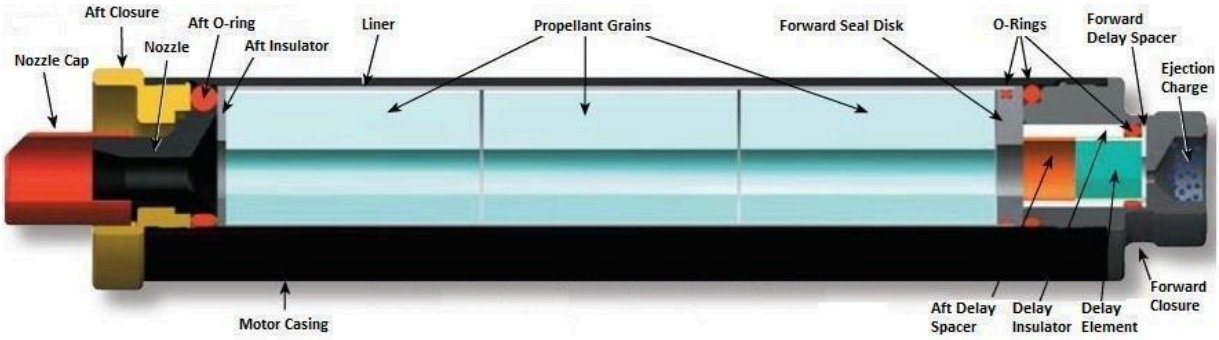


Figure 6-3: Cutaway Drawing of Typical Assembled RMS-Plus Motor

6c. E-Matches / Motor Ignitor

Electric matches, commonly called "*E-Matches,*" are a universal initiator of many rocketry pyrotechnics and motors. A typical e-match is made from a thin nichrome (nickel-chromium) wire laminated to a small non-conductive flake of fiberglass. Each end is soldered to one wire of a two-conductor solid core copper shooter wire. The nichrome bridge is dipped into a pyrogen solution that dries hard and looks like a match, hence the name.

Figure 6-4: Aerotech Single Use Motor with Ignitor

E-matches are typically high current or low current. Kits for making your own e-matches can be purchased on the internet, or you can purchase them from most vendors on launch day.
E-matches can be augmented to serve as motor igniters also.

6d. Thrust to Weight Ratio

In order for a rocket motor to lift a rocket, it must produce enough thrust to overcome the force of gravity. This means a rocket motor, at a minimum, must produce enough mechanical energy to achieve a *thrust-to-weight ratio* of just over 1.0. For example, if your rocket has a thrust to weight of 5:1, that means the rocket motor must produce 5 times the force of the weight of your rocket. If your rocket weighs 20 lbs. then the motor needs to produce at least 100 lbs. of thrust. http://www.thrustcurve.org/

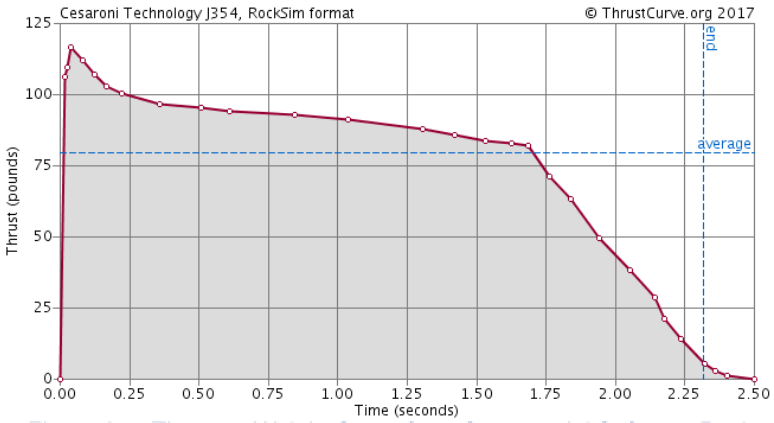


Figure 6-5: Thrust to Weight Curve from Commercial Software Package

6e. Trajectory Analysis

Below is an example of a completed trajectory analysis. The analysis displays important events in the flight such as maximum altitude, velocity, and acceleration. It may also display the rocket's recovery events and descent rate.

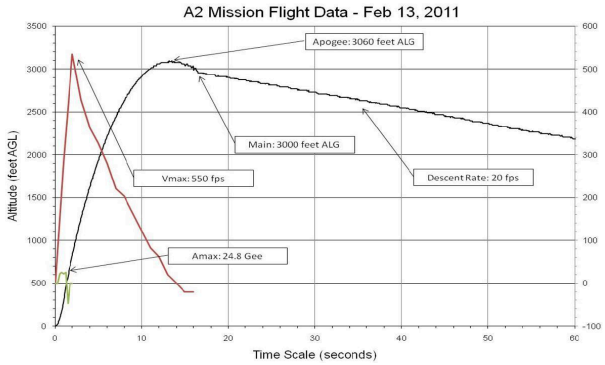


Figure 6-6: Plotted Trajectory Data from Commercial Software Package

6f. Commercial Software Packages

The easiest way to perform a trajectory analysis is to purchase a commercially produced trajectory analysis program like RockSim, or RASAero. There are also free software packages available like OpenRocket. These programs have simple computer-aided design (CAD) programs that allow a user to build a model of the rocket in the program. The programs can help calculate many rocket parameters like weight, length, and stability in addition to performing a three degrees of freedom (3DOF) flight analysis.

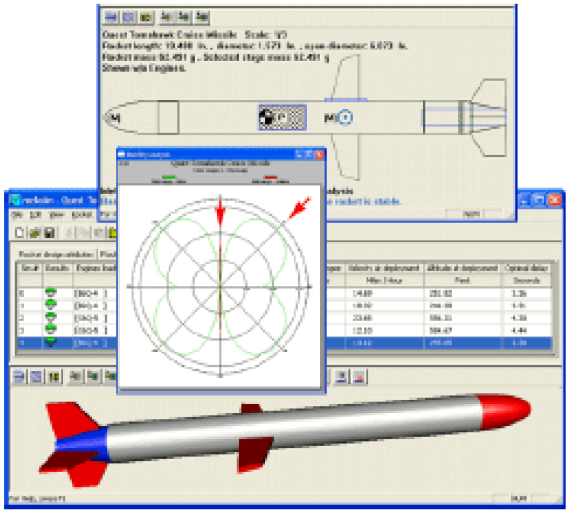


Figure 6-7: RockSim Flight Simulation Software

These commercial programs offer the user a lot of control over the simulation environment also. The user can manipulate the winds, the launch angle, temperature, and many other parameters to match the simulation environment to the real world.

Commercial software can be found at the following links:

- RockSim: http://www.apogeerockets.com/rocksim.asp
- RASAero: http://rasaero.com/
- OpenRocket: http://openrocket.info

Submission of RockSim file simulation is required for all WSGC rocket programs.

6g. Propulsion System Notes

7. Stability

An important component of rocket vehicle design, and a safe and successful flight, is ensuring vehicle stability.

An object is directionally stable if it tends to return to its original direction in relation to the oncoming medium (water, air, etc.) when disturbed away from that direction. Directional stability is also called “weathervaning or weathercocking” because a directionally stable vehicle free to rotate about its center of mass is similar to a weather vane rotating about its pivot. Without stability, a rocket would tumble end over end, spin, or orient itself at a high displacement angle. At high displacement angle, drag forces may become excessive and the rocket may experience structural failure.

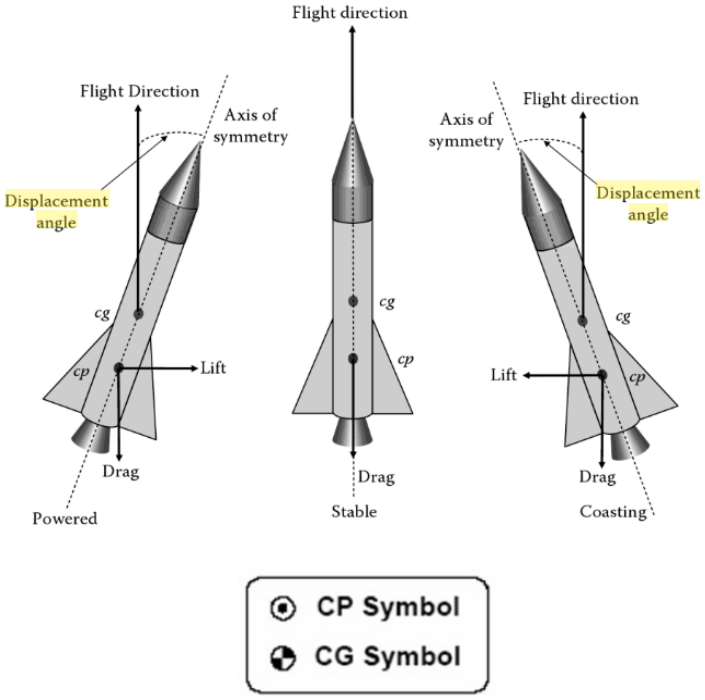


Figure 7-1: Rocket Stability Diagram

Generally, a rocket is considered stable if its center of gravity (CG) is at least one body diameter in front of its center of pressure (CP). A practical approach to efficient rocket design is to allow the structural design to mature to the point where the CG location is stable (does not vary much with tweaks to the configuration) and then tailored for the desired stability margin by selecting where you want your CP to be located.

7a. Center of Gravity

The *center of gravity (CG or* $\bar{X}_{CG}$*)* of a rigid body is the mean location of all the masses in a system.

The position of the CG is fixed in relation to the body and does not coincide with the geometric center generally. The CG can be determined analytically or empirically. The standard CG symbol in Figure 7-1 should be added to the rocket via stickers or pen.

The *Analytical Approach* requires accounting for all of the individual point masses that compose the system and their location in the system as measured from a common datum plane, typically the tip of a rocket's nose cone. The average of their positions weighted by their masses is the location of the center of gravity. The basic assumptions used in calculation of the theoretical center of gravity for this rocket are uniform gravitational field (g = constant) and that the components have uniform density (ρ= constant). Simulation programs, such as RockSim, would use this method.

The *Empirical Approach* relies on observation and experience. An example of determining the CG empirically would be a simple balance method. Find the point on the rocket where it balances and you have found the CG. Although accurate, this method is not practical on very large and heavy rockets and is not useful during the design phase of your student launch project. It can be an easy check just to verify your analytical model once your rocket is complete.

7b. Center of Pressure *(Build reference 13.b.viii)*

As a rocket flies through the air, aerodynamic forces act on all parts of the rocket. In the same way that the weight of all the rocket components acts through the center of gravity (CG), the aerodynamic forces act through a single point called the *center of pressure (CP or* $\bar{X}_{CP}$). The standard CP symbol in Figure 7-1 should be added to the rocket via stickers or pen.

You can calculate the CP, but this is a complicated procedure requiring the use of calculus. The aerodynamic forces are the result of pressure variations around the surface of the rocket. In general, you must determine the integral of the pressure times the unit normal, times the area, times the distance from a reference line. Then divide by the integral of the pressure times the unit normal, times the area. Lots of work!

A simple empirical method is the cardboard cutout method. This method assumes that the center of pressure coincides with the centroid, or geometric center, of the rocket. Make a cardboard cutout of the rocket silhouette and find the balance point. This method is an easy approximation of the area where the CP might be located. This method could be useful early on in the preliminary design of your student launch projects, but is not recommended as your primary method for determining your rocket's CP.

7c. Static Margin

Static margin (SM) or *margin of stability* describes the directional stability of a rocket. Recall that an object is directionally stable if it tends to return to its original direction in relation to the oncoming medium (water, air, etc.) when disturbed away from that direction and that a rocket is considered stable if its *CG* is at least one body diameter in front of its *CP.*

$$SM = \frac{\bar{X}_{CP} - \bar{X}_{CG}}{Body\ Diameter} \geq 1.0$$

Equation 7-1: Static Margin

Generally, it is desirable to have a static margin of 1.5 to 2.0. A rocket is considered over-stable if it has a static margin of 3.0 or greater. An over-stable rocket will lean or "weather vane" further into the wind and not travel as high. It is important to note that a rocket's CG will change as the motor exhausts combusted fuel. Generally, the CG will move forward as a solid rocket motor burns, causing the rocket to become more stable.

7d. Stability Notes

8. Recovery Hardware *(Build references 13.b.v, xii)*

High-powered rockets are required to have a recovery system. In this section, you will learn about the hardware needed to build a recovery system, how to apply Newton's 2nd Law of Motion to calculate the size parachute needed to safely recover your rocket and how to use the *Ideal Gas Law* to appropriately size black powder ejection charges to deploy your recovery systems.

High-powered rockets typically have one of two types of recovery systems - *Single Event Recovery* or *Dual Event Recovery*.

8a. Single Event Recovery System (SERS)

A typical S*ingle Event Recovery System* ejects a parachute at apogee. This can most commonly be achieved by using a motor ejection charge. Most low- to mid-powered rocket motors have this capability. At motor ignition, the propellant and a delay grain begin to burn. The delay grain burns slowly. Once it burns through, the ejection charge is set off and the parachute is deployed. If the timing is good, this happens near apogee. SERS is the simplest recovery system and is good for low altitude flights on small launch fields and high-altitude flights on large launch fields.

Figure 8-1: Single Event Recovery System

The plot shows the trajectory of a rocket with a single event recovery system. The rocket reaches apogee and deploys a parachute using the motor ejection charge. The rocket then descends slowly at 20 fps. With an apogee of 3000 feet, it takes two and a half minutes for the rocket to touch down.

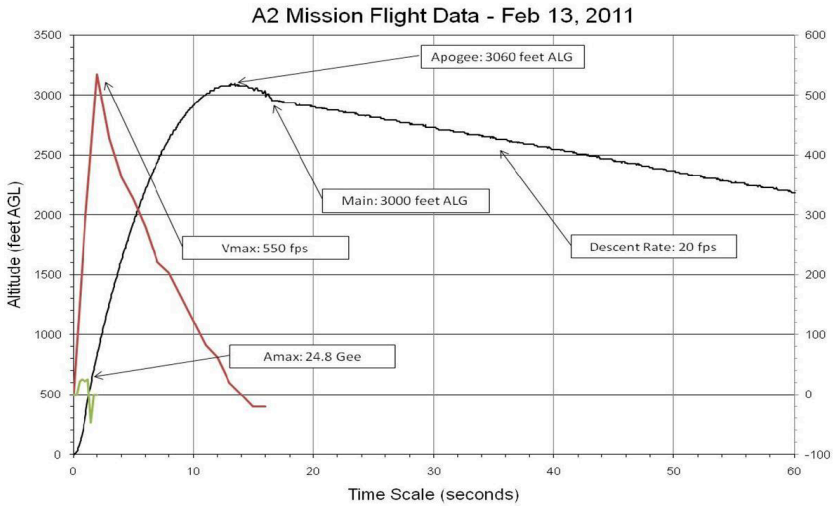


Figure 8-2: Single Event Trajectory Plot

8b. Dual Event Recovery System (DERS)

A typical *Dual Event Recovery System* has one event at apogee and the second at a much lower altitude, typically 700 feet or more *Above Ground Level (AGL)*, and requires electronics to do so. This recovery technique significantly reduces the recovery area by allowing the rocket to fall much faster from apogee and deploying a main parachute much closer to the ground. The first event recovery system is typically a long length of shock cord and perhaps a drogue parachute. The second event recovery system is the main parachute, which slows the rocket down considerably for a safe touchdown.

Figure 8-3: DERS Example

The following plot shows the trajectory of a rocket using a dual event recovery system. The rocket reached an apogee of 4800 feet. It then deployed a drogue that slowed the descent to 90 fps. At 1100 feet, the main parachute deployed further slowing the descent rate to 18 fps. It takes only one-and-a-half minutes for the rocket to touch down.

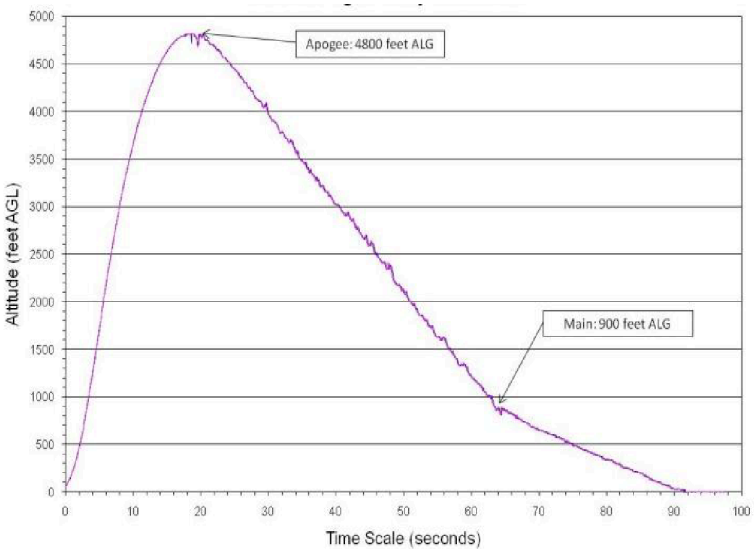


Figure 8-4: Dual Deploy Trajectory Plot

A rocket equipped with a typical dual event recovery system has the following general layout. The main parachute is generally forward of the avionics bay. This adds advantage to the rocket's center of gravity and hence the rocket's stability margin by having the larger and heavier of the recovery devices far forward.

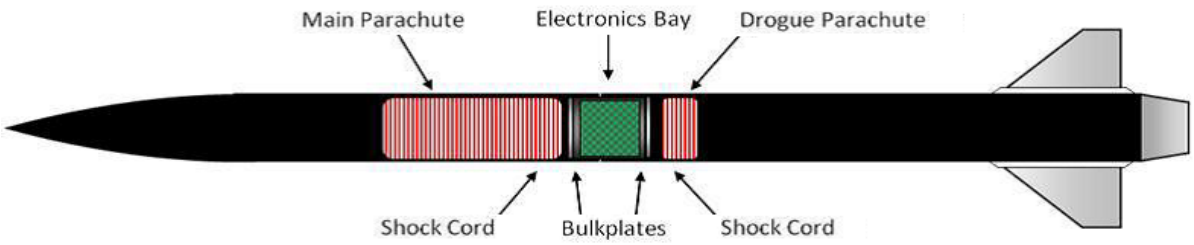


Figure 8-5: Possible Layout of a Rocket with DERS

The avionics bay is generally between the two parachutes. This adds advantage to the locations of the recovery ejection charges. The charges are generally placed in cups on the avionics bay's end closures or bulkplates. Avionics bays are discussed further in Section 9.

8c. Parachutes

Parachutes are the most commonly used recovery device in high-powered rocketry. A parachute is a high drag device that reduces the high-speed descent of the rocket by producing a force that opposes the weight of the rocket. The effectiveness of a parachute depends on velocity, air density, surface or "reference" area, and a drag coefficient. A drag coefficient is a dimensionless quantity that is used to quantify the drag of resistance of an object in a fluid environment such as air. For non-streamlined objects like parachutes, the drag coefficient can be greater than 1.

Figure 8-6: Dual Deploy Recovery System

High-powered rocketry parachutes typically have three major features:

1. Canopy
2. Shroud lines
3. Steel connector link

The canopy is made of several rip-stop nylon cloth panels, or gores, sewn together to form a round, cruciform, annular, or other shape. The shroud lines are attached to the outer edge of the canopy. The steel connector link gathers all the shroud lines together and can be connected to the rocket's recovery harness. Some of these connectors also have a swivel built in to aid in stability. Some parachutes have a spill hole or apex vent - a small hole in the top of the canopy that allows small amounts of air to spill out from the top of the canopy adding to its stability. Otherwise, the parachute will rock from side to side to dump air out from the bottom sides of the canopy, which would cause the rocket to swing like a pendulum.

8d. Parachute Selection

There are two criteria to assist with parachute selection. The descent velocity, or ground-hit velocity, should be at or under 20 ft/s for minimal body damage upon landing. The expected drift range should also not exceed 2500 feet downrange. Most rocket simulators will determine the descent velocity for you if the model is accurate.

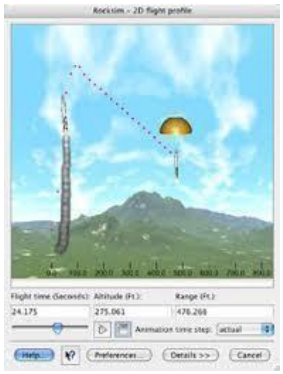


Figure 8-7: Parachute Selection with Simulation Data

8e. Parachute Protectors

Nylon cloth is susceptible to melting and charring. It is necessary to protect your recovery system from the hot gases and any burning debris that are generated by the ejection charge. A flameproof Aramid cloth or Kevlar cloth will protect the parachute from these hot gases and burning debris. Other options include deployment free bags (D-bags). D-bags provide parachute and support line protection and can ensure a reliable and orderly deployment. Another option is to utilize wadding (dog barf) when packing the parachute. Great care should be taken to not tightly wrap the parachute with both techniques.

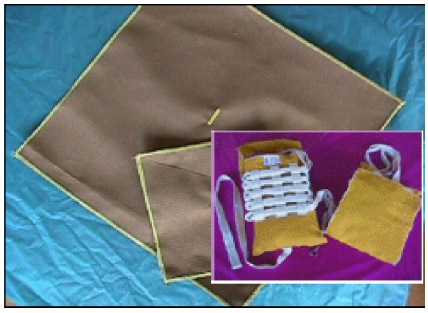


Figure 8-8: Parachute Protector

8f. Recovery Hardware

A *recovery harness* is a generally long length of static cord that attaches structural components that separate as part of the recovery system. Typical materials for a recovery harness are Aramid/Kevlar, Kevlar/Fiberglass, and nylon. Each end is secured by a quicklink typically to a *U-bolt* or *eyebolt* that is rigidly mounted to a bulkplate on an avionics bay, nose cone, or booster. The main parachute is generally attached to a loop in the shock cord tied just below the nose cone by a quicklink. The length of recovery harness required will be the subject or empirical testing and evaluation. Your workshop rocket has 12 feet of shock cord. A Level 3 type rocket may require more than 100 feet of shock cord.

Figure 8-9: Recovery Hardware

The recovery harness can be bundled into groupings using rubber bands or masking tape. This technique helps to dissipate the momentum of the separating components when they are ejected. In the photo above, the shock cord was bundled into four 10 feet long lengths that were

Figure 8-10: Recovery Harness

folded in 8” - 10” lengths and wrapped a couple of times with masking tape. During deployment, energy is used up to tear the tape to release more cord. This technique performed very well in-flight dissipating momentum of the two 25 lbm rocket sections that were ejected at 200 lbf before the shock cord became taught. Recall that nylon is susceptible to melting and charring. Take measures to protect nylon cords and inspect all shock cords before flights.

8g. Quicklinks / Eyebolts

Quicklinks make connecting and rigging a recovery system very easy. There are many different types of quicklinks available commercially. Always use quicklinks with a locking gate. In addition, the by-products of black powder ejection charges are acidic. While zinc-plated steel quicklinks are safe for many flights, inspect the levels of corrosion and clean all metallic hardware between flights. Take care in choosing a quicklink that will safely carry the maximum expected load that will be experienced. Too small of a quicklink may yield under heavy loading when the main parachute opens.

The quicklinks attach the recovery system to the rocket’s structures. There are several hard points on the rocket’s bulkplates and/or centering rings. These hard points are generally made using eyebolts or U-bolts. U-bolts are preferred on large rockets. Eyebolts should be closed and/or welded closed.

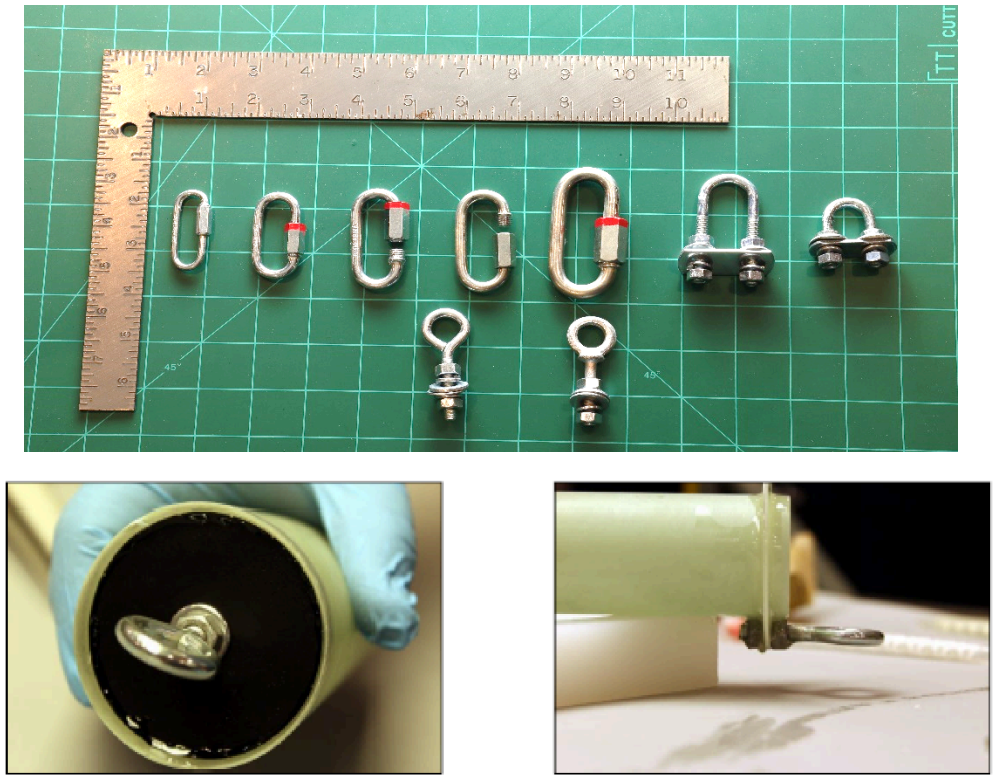


Figure 8-11: Typical quick links, U-Bolts and Eyebolts (Upper Left); Quicklinks (Upper Right); Closed Eyebolts (Bottom)

8h. Recovery Notes

9. Avionics System

Every dual deploy rocket will require recovery electronics, to control the ejection charges for the recovery events. Recovery electronics are optional for single deploy rockets, where the ejection charge is usually contained within the motor, and is on a timed fuse.

Recovery electronics will usually be contained within the electronics (avionics) bay but can be in other locations within the rocket. Recovery electronics also cover the tracking electronics that are used to locate a rocket that lands over the horizon (out of sight).

9a. Avionics Bay

Avionics refers to any electronic systems flown on a rocket, whether they are flight computers, guidance and control systems, telemetry systems, or payloads. These systems are typically built into an Electronics Bay. An 'E-Bay' is a subsystem of a high-powered rocket that typically contains altimeters, batteries, and switches. A typical E-Bay consists of three structural components: the housing, a forward and aft endcap, and an avionics sled.

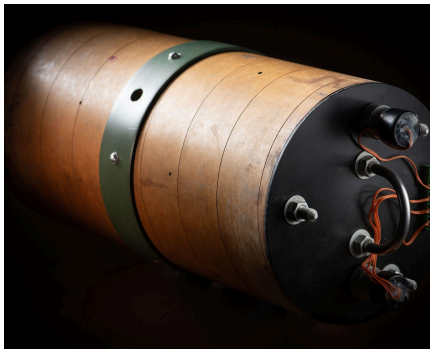


Figure 9:1- Avionics Bay

9b. Avionics Bay Housing / Coupler

The E-Bay housing is typically built from a coupler tube and can have a collar made from a segment of airframe that is epoxied to the middle of the coupler and allows direct access to the switches that power on the altimeters. The collar also supports static pressure ports that equalize the housing's interior pressure with the exterior atmosphere. Generally, rocket builders follow the convention that when joining airframes with a coupler, the coupler should extend at least one airframe diameter into each joined segment.

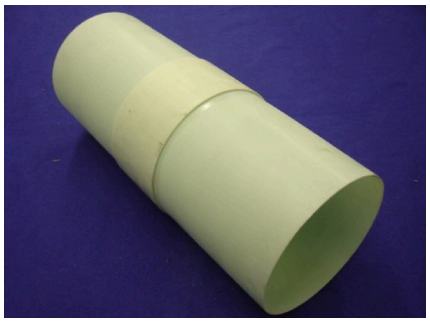


Figure 9-2: Coupler Tube

9c. End Caps

The E-Bay end-caps close out the housing, separating the rocket's volumes, and support the recovery harness hard mounts, charge cups, and all-threads. Typically, end-caps are made from G-10 fiberglass bulkplates, or plywood. The all-threads act as a two-force member that connect both end-caps and carry the recovery harness load through the E-Bay. End-caps should create a good seal around the end of the housing to prevent hot gas seepage from the ejection charges.

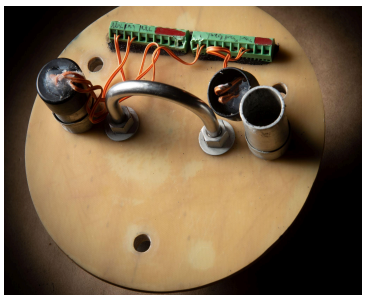


Figure 9-3: End Cap/Bulkplate

9d. Avionics Sled

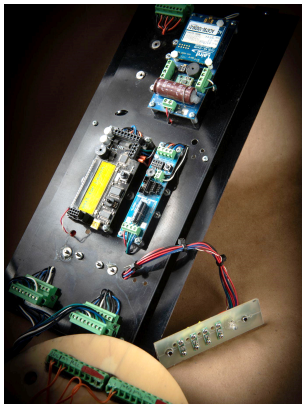


Figure 9-4: Avionics Sled on G-10 Fiberglass Tubes

Typically, the avionics sled is a G-10 fiberglass board or boards that mount in the avionics by sliding onto the threads that connect the end plates, like a sled. The avionics electronics, batteries, and switches are mounted to the sled and wired together to form systems. The avionics sled in the photo to the right is slid onto ¼" diameter all-threads using ¼" I.D. G-10 fiberglass tubes that are epoxied to the corners on the G-10 boards. This simple sled supports only altimeters, batteries, and switches. The sled was designed to slide into the avionics bays of several different rockets.

The system level diagram below details the dual event recovery system's configuration. This system is a fully redundant recovery system. There are two altimeters, each have a dedicated battery and switch. Each altimeter has its own set of recovery charges to fire. There are four in total in the rocket. Although not required, redundant setup is a nice feature.

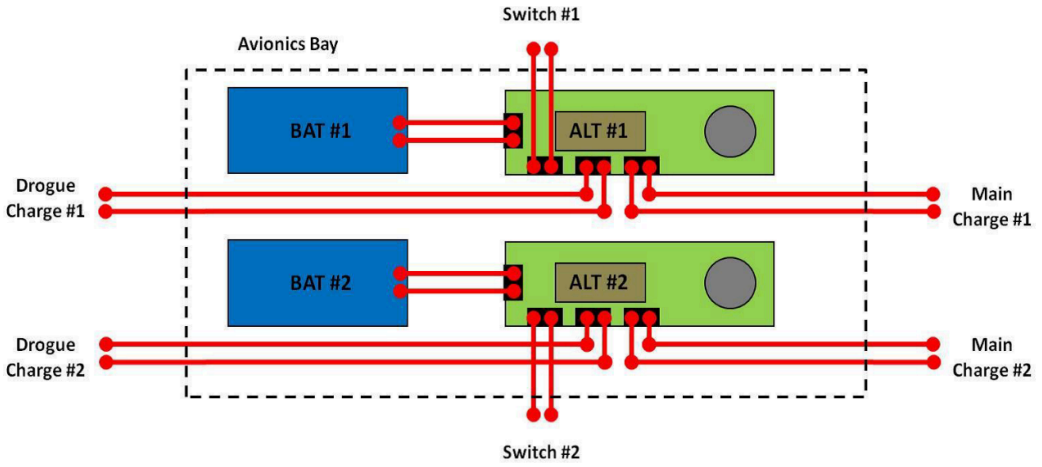


Figure 9-5: System Level Diagram of DERS Configuration

9e. Recovery Electronics

Dual Event Recovery Systems require the use of electronic devices called *Altimeters* that can determine altitude and initiate events at desired altitudes.

Altimeters, usually mounted in the rocket's avionics bay, need to sample the outside air pressure. Your rocket will need a static pressure port along the outside to allow the inside pressure to equalize to the outside pressure.

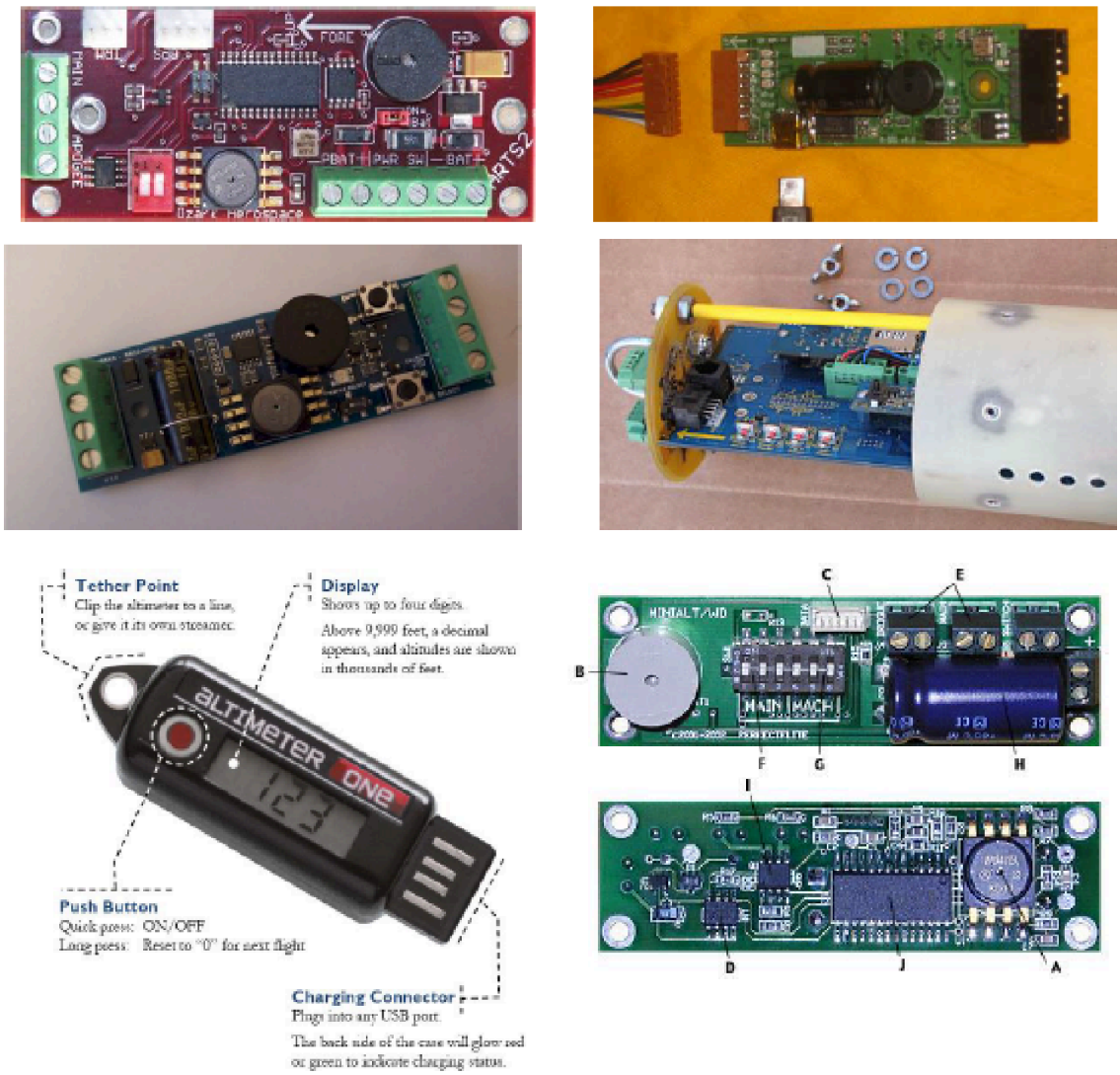


Figure 9-6 Clockwise from top left: RRC2 Mini Alt, G-WIZ Av Bay, AED R-DAS Tiny, Perfectflite MAWD, Apogee Altimeter One, ARTS2 Flight Computer

9f. Batteries

Typically, rocket builders prefer each altimeter to have a dedicated battery. Your electronics should be able to remain switched on and reliably operate if the rocket remains on the pad for an extended duration before launch.

There are two main types of batteries used in high-powered rocketry, primary and secondary. Primary batteries such as a 9V alkaline battery are designed for one-time use and then discarded when they are exhausted. Even if never taken out of the original package, primary batteries can lose 8% - 20% of their original charge every year when stored at room temperature. This "self-discharge" rate is known to occur due to a non-current producing side chemical reaction, which occurs within the cell even if no load is applied.

Secondary batteries, like Nickel Cadmium (NiCad) or Lithium Polymer (Li-Po) batteries are designed to be rechargeable and used multiple times. Secondary batteries weigh less than primary batteries and manufacturers can shape them; however, they please. They are more expensive than primary batteries. Some require sophisticated chargers to recharge them safely. Improper use or charging of some secondary batteries can result in fire or explosion.

Secondary batteries self-discharge more rapidly than primary batteries. A freshly charged NiCad battery can lose 10% of its initial charge in 24 hours, and discharges at a rate of about 10% every month thereafter. Most Li-Po batteries have reduced self-discharge-rates to a relatively low level but are still poorer than primary batteries. Even though secondary batteries have their energy content restored by charging, some deterioration occurs on each charge/discharge cycle. Secondary batteries like Li-Po batteries are gaining favor in the work of high-powered rocketry where the advantages of both lower weight and greatly increased run times can be sufficient justification for the price.

9g. Switches

Typically, rocket builders prefer to power up their electronics from the outside of the rocket once they are placed on the launch pad. This method maximizes battery life. The switches can either be surface mounted to the airframe of the rocket or mounted inside the rocket with an access hole or panel. The two categories of switches used on high-powered rockets are the *Single Pole Single Throw (SPST)* and the *Dual Pole Single Throw (DPST)*. The *Single Pole Single Throw (SPST)* is a simple on-off switch where the two terminals are either connected or disconnected from each other.

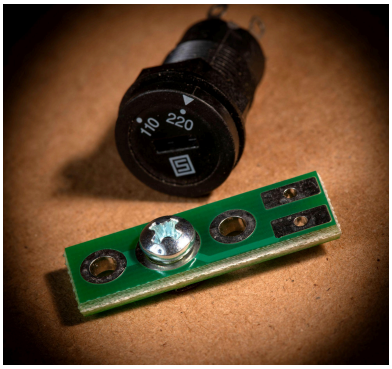


Figure 9-7: Single Pole Single Throw (Top); Dual Pole Single Throw (Bottom)

The *Dual Pole Single Throw (DPST)* is equivalent to two SPST switches controlled by a single mechanism. In these two categories, there are toggle switches, push button switches, and selector switches.

9h. Tracking

There are three main types of tracking: audible, GPS, and RF.

In general, FCC certification is required for any electronic device that can oscillate above 9 kHz. Manufacturers must make sure that their products will not interfere with other products or cause risk and harm to the public. All electrical devices must be tested and fulfil the emission requirements to receive FCC certification. If manufacturers sell products without the appropriate approval, they will receive a monetary fine and their products will be recalled.

Teams purchasing a tracker must be aware if the tracking unit requires an FCC certificate or equivalent. The manufacturer or seller should abide by these rules also and require that the consumer has a certificate or will acquire one.

For introductory rocketry, it is recommended to use an electronic tracking device that does not require licensing.

i. Audible

This is the cheapest and simplest option. There are commercial products produced for high-powered rocketry that emit a high-pitched audible sound that are activated when the parachute is ejected from the rocket. The sound will help you to locate the rocket, but the range is not very large, so you must know the general vicinity of the rocket upon landing.

ii. RF Tracking

This is an option to assist with rocket recovery. There are commercial products for high-powered rocketry, which produce an RF pulse on a distinct frequency. The device is installed in the rocket, and using a Yagi antenna tuned to the frequency, the rocket can be homed in on.

iii. GPS Tracking

This is the most expensive option but is the easiest and most reliable. Again, there are commercial products for high-powered rocketry, which emit GPS coordinates (position) in real time to a receiver.

9i. Ejection Charges

A black powder charge is the most common and reliable method of ejecting a parachute from your rocket. In your rocket, the motor's ejection charge will ignite and generate hot gases that pressurize the rocket's airframe and exert a net force on the bulkplate of the nose cone. This net force will eject the nose cone, shock cord, and parachute out of the rocket airframe. This all happens because the rocket is obeying the *Ideal Gas Law*.

The *Ideal Gas Law* is the equation of state for a hypothetical incompressible or "ideal" gas. The state of the amount of gas is determined by its pressure, volume, and temperature.

The modern form of this equation is PV = NRT where *P* is the absolute pressure of the gas, *V* is the volume occupied by the gas, *N* is the amount of substance (in this case the substance is black powder), *R* is the gas constant, and T is the absolute temperature. The equation can be reordered to solve for N directly and known values substituted. The design pressure is determined by the desired net force on a surface divided by the area of that surface. Typical net force values for a 4-inch diameter rocket range from 100 lbf - 200 lbf. This translates to a typical pressure range of 8psi – 16psi. In addition, black powder charge amounts are typically reported in unit grams. Recall there are 454 grams in 1 pound. So, for a rocket with r = 2 inch:

$$P = \frac{F}{A} = \frac{F}{\pi r^2} = \frac{200\ lbf}{\pi(2\ in)^2} = 16\ psi$$

Equation 9-1: Pressure

$$V = \pi r^2 L = 12.56\ in^2 \cdot L$$

Equation 9-2: Volume

The volume is a function of the length of the rocket tube being deployed (left as a variable).

$$N = \frac{PV}{RT} = \frac{(16\ psi)(12.56\ in^2 \cdot L)}{\left(266\ \frac{in \cdot lbf}{lbm}\right)(3307°R)}\left(\frac{454\ grams}{1\ lbf}\right) = (0.1 \cdot L)\ gram$$

Equation 9-3: Black Powder

We see that the reduced equation for this case states that 0.1 grams of black powder is needed for every 1-inch length of airframe containing the recovery system. Therefore, if L = 20 in, then 2.0 grams of black powder are needed to eject the recovery system with 200 lbs. of force. This theoretical value should now be tested.

Figure 9-8: Ejection Charge Cup

Typically, rocket builders use a charge cup or well to contain the measured amount of black powder for the ejection charges. Everything from PVC end caps, brass pipefittings, to a rolled length of blue masking tape can be used. Some fixed volume charge cups will require a filler of some type to occupy the empty volume in the cup once the black powder and e-match have been installed. Rocket builders typically use soft foam earplugs or shredded housing insulation material more commonly called “dog barf.” Cannon plug covers, electrical tape, masking tape, and duct tape can be used to seal the charge cups.

Alternatively, ejection charges can be contained in a prefabricated plastic container (canister), with a built-in filament to ignite the charge. These containers will come in small and large sizes, and a foam plug needs to be used to fill the remaining empty space in the container, so that the black powder remains in contact with the filament at the bottom of the container. Wire leads extend from this ejection canister, which are then simply wired to your wiring block on the outside of your avionics bay, or can be wired directly to the altimeter. Care must be taken to direct the charge along the length of the rocket airframe tube.

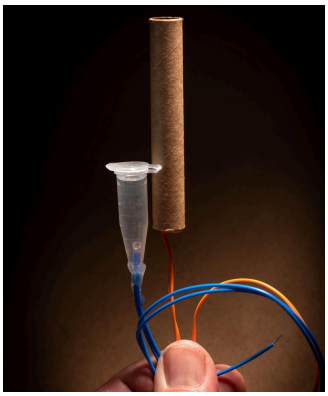


Figure 9-9: Ejection Canister

9j. Avionics Notes

10. Recovery Exercise: Ejection Charge Sizing

A black powder charge is the most common and reliable method of ejecting a parachute from your rocket. This exercise will teach you how to determine the ideal amount needed for your rocket by using the *Ideal Gas Law*.

$$PV = NRT$$

Equation 10-1: Ideal Gas Law

$$P = \frac{F}{A}$$

Equation 10-2: Pressure

The constants for 4F black powder are:

$$R = 266\ \frac{in \cdot lbf}{lbm}\ (Gas\ Constant) \qquad T = 3307°R\ (Temperature - Rankine)$$

The variables for your rocket are:

$$F = 100\ lbf\ (Ejection\ Force)$$

$$A = 5.0\ in^2\ (Area)$$

$$V = 100\ in^3\ (Volume)$$

Use the *Ideal Gas Law* and Pressure formula to determine the required amount of black powder to eject a parachute from your rocket. **Note: There are 454 grams in 1 lb.**

10a. Ejection Charge Calculation Worksheet

11. RockSim Overview

11a. Creating a Model

Simulations are THE key component to high-powered rocket design. Models can be created of an existing rocket in RockSim. We will create a simple model of your as-built LOC Precision Launch 2 Learn rocket.

- RockSim: http://www.apogeerockets.com/rocksim.asp

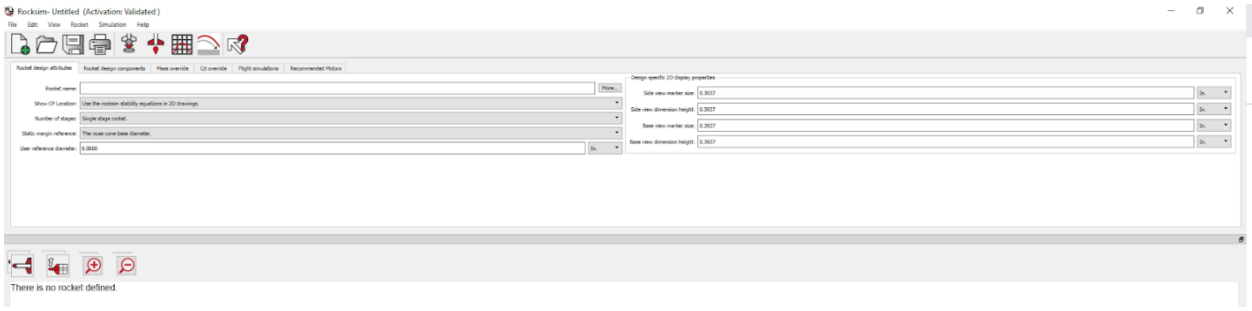


Figure 11-1: RockSim 11

Learning to use RockSim

- Rocket Design Attributes
- Rocket Design Components
- Mass Override
- CD Override
- Flight Simulations
- Recommended Motors

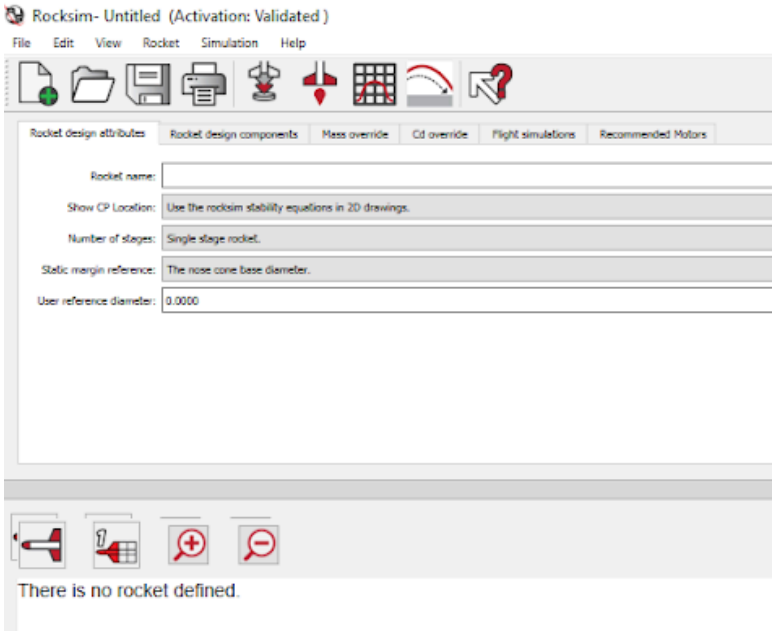


Figure 11-2: RockSim 11 Folders

11b. Importing a Model

Models of kit rockets can be imported into RockSim. This feature is useful to perform simple analysis to ensure the kit is suitable for the competition parameters. It is quick to import a model and run a simulation of that model. Most rocket vendors/suppliers will offer a free RockSim model, such that you can perform analysis first, prior to purchasing the kit. We will search for and import a detailed model of the LOC Precision Caliber-ISP. There are databases online, where users can upload and share or download RockSim models.

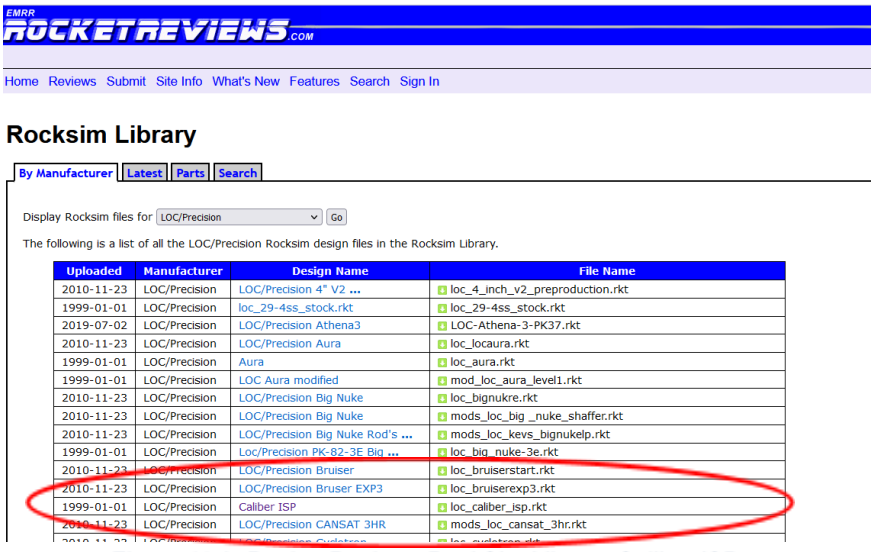


Figure 11-3: Rocket Reviews RockSim Library, Caliber ISP

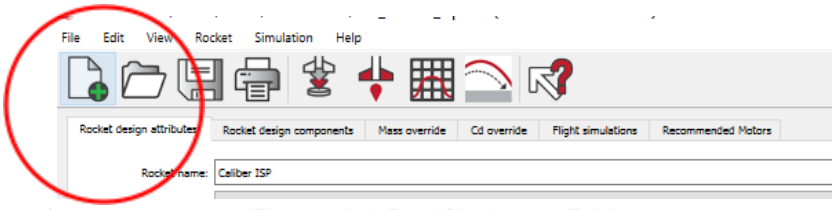


Figure 11-4: RockSim Import Folder

i. Selecting a Kit

1. Download a pre-existing model of a LOC Precision-Caliber ISP kit (The L2L kit is based off of the pre-existing model and requires modifications)
   a. https://www.rocketreviews.com/rocksim-library.html
   b. Sort by Manufacturer
   c. Display RockSim files for LOC/Precision
   d. Download Caliber ISP file
0. Import rocket file into RockSim 11
   a. Open the existing design file folder (Import Folder)
   b. Select loc_caliber_isp file from download folder
3. View the LOC Caliber ISP

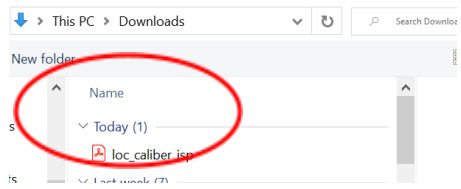


Figure 11-5: Select Rocket File from Download Folder on Computer

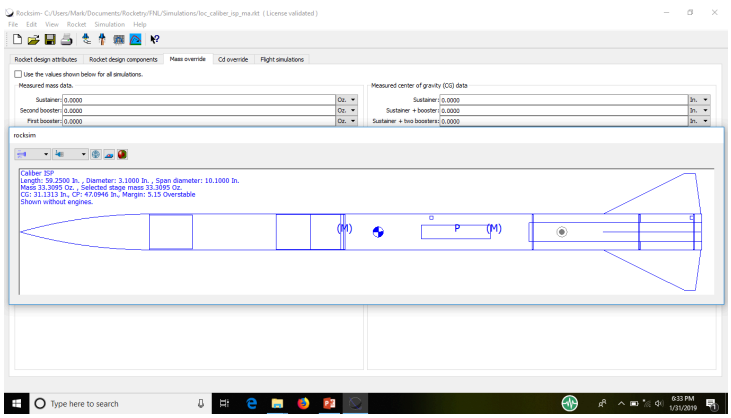


Figure 11-6: Caliber ISP Stock File

ii. Selecting an Engine

1. Select the “Prepare for Launch” icon

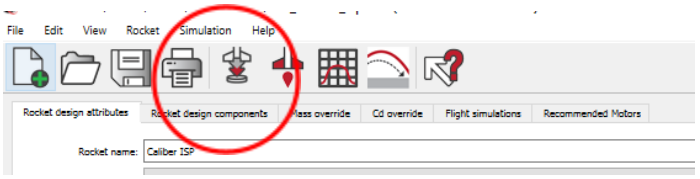


Figure 11-7: Select Prepare for Launch

2. Select the “Choose Engine” tab

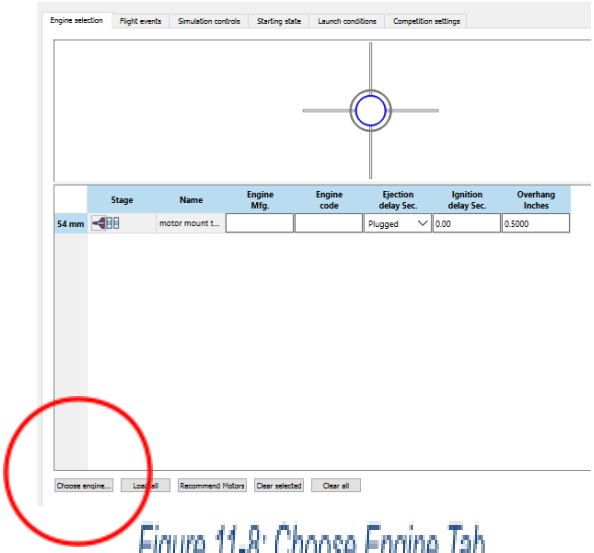


Figure 11-8: Choose Engine Tab

3. Select “Aerotech” in the Manufacturer Filter field

4. Select the “Show all engines” in the Diameter Filter field

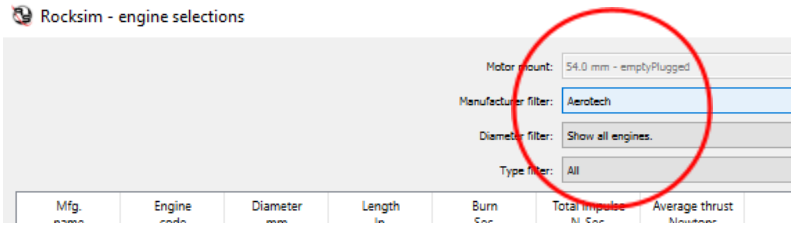


Figure 11-9: Manufacturer Field and Diameter Filter

5. Double click on the H100W motor

The AeroTech H100W will automatically load into the RockSim software.

Mfg. name	Engine code	Diameter mm	Length In.	Burn Sec.	Total impulse N-Sec.
Aerotech	G79W	29.00	5.9000	1.42	107.054
Aerotech	G8	29.00	6.0236	17.27	138.181
Aerotech	G80NBT	29.00	4.8752	1.70	135.628
Aerotech	G80T	29.00	4.8819	1.50	116.252
Aerotech	G85T	29.00	6.1417	19.96	129.868
Aerotech	H100W	38.00	6.2008	2.40	228.405
erotech	H112J	38.00	7.5197	3.40	280.685
erotech	H115DM-14A	29.00	7.9921	1.69	177.718
erotech	H123W	38.00	6.0630	2.60	211.400
erotech	H123W	38.00	6.0630	2.45	235.175
erotech	H128W	29.00	7.6772	1.50	155.795
erotech	H128W	29.00	7.6772	1.50	155.764
erotech	H130W	38.00	5.9843	1.63	210.221

Figure 11-10: Aerotech H100WMotor Selection

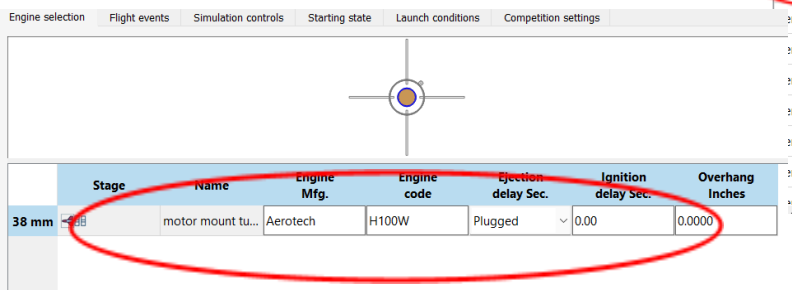


Figure 11-11: H100W Motor Uploaded to RockSim

11c. Running a Flight Simulation

Once a model is created or imported, RockSim can be configured to perform simulations of your rocket, under various conditions, with various motors. We will set up and perform a flight simulation of the LOC Launch 2 Learn rocket kit..

11d. Interpreting Results

Interpreting and understanding the results of a flight simulation are important, to ensure requirements are satisfied. If the results do not satisfy requirements, then modifications to either the simulation conditions, the motor selection or the parachute selection should be made, and the flight simulation repeated. If the results still do not satisfy requirements, then perhaps a different kit/model should be imported and the whole process repeated.

Interpreting the results properly will also ensure that the simulation was set up and performed properly.

11e. Simulation Tips

In the proposal and preliminary phase, many simulations should be run, examining as many kits as possible, to determine the best kit-motor-parachute combination that will satisfy requirements. By the time of the submission of the PDR, the team should have selected one kit-motor-parachute combination.

Once the kit arrives, and the build progresses, keep in mind that modification will most likely be made to the stock kit, to accommodate your challenge payload/solution. Simulations must be updated to match the existing rocket, to ensure any changes will not violate vehicle performance requirements.

11f. RockSim Notes

12. Designing and Acquiring Your Rocket

As a new rocket builder, designing and building your first high-powered rocket for competition may seem like a daunting task. This section will highlight some of the key components to help you with a safe and successful flight.

12a. Rocket Selection / Rocket Simulation

The first step is selecting which kit you will purchase to build your competition rocket. To be successful, it is beneficial to view the widest array of kits possible, so you know what's out there. Selection of the kit is not trivial; be sure to spend some time researching the capabilities and allow the payload design to mature.

12b. Simulations

Simulating the flight performance of the kit, on the expected motor (or similar) is beneficial prior to buying your kit. Usually, vendors and manufacturers will provide a RockSim file such that you can simulate accurately. A range of kit and motor combinations should be considered. Once the kit is narrowed down and selected, it is a good idea to have a backup motor selection in mind, as the weight increases on your rocket over the course of the design/build.

In conjunction with initial simulations to help select a kit, payload design should begin. Discuss concepts and ideas, while looking at cost, weight, and size. The rocket kit should not be selected until it is determined the maximum size of the payload that will fit inside (aka: don't let a random kit diameter choice dictate how small your payload should be).

Conversely, the estimated payload weight should be known as early as possible (or an upper limit set), as this will also determine the motor choice, since this estimated weight will need to be added to the stock simulations and affect the altitude requirement.

12c. Weight (Simulation vs. Stock Kit vs. Modified Kit with Payload)

It is usually a rule of thumb that initial simulations will under predict the final weight of the fully loaded rocket. Drag considerations will also likely be under predicted. This should lead to an initial higher than required altitude estimate (or stated differently, the initial predicted altitude can be conservatively high, with the assumption that weight will increase by final design). If the weight increase is too much, then an increase in motor size may also have to occur.

Keep in mind there is a deadline for motor selection, after which you can no longer change the motor you will be using for competition flight!

12d. Recovery

Determining the size of the parachute and the size and how much recovery hardware should be completed prior to the kit is purchased. The size of the parachutes is not trivial as they will need to satisfy descent rate requirements and will add weight to the rocket vehicle.

12e. Miscellaneous Design Tips

Don’t just randomly buy a kit too early because it says it reaches a certain altitude! A stock kit can likely be assembled within a week by a team of students. A modified kit may take an extra week or so. If you want to be built by FRR, then set aside all of March to build, and that should be plenty of time. If your design looks good at PDR, you can begin ordering parts in February.

Be sure to understand how changes to the design affect performance. For example, if your build weight increases to the point where you need to choose a bigger motor, this will in turn add more weight toward the rear of your rocket, which will change your stability margin (CG vs. CP) thus, weight may have to be added towards the nose of the rocket to satisfy the required stability margins.

Changing to a bigger motor will also increase the overall weight, which will increase the descent rates and kinetic energy upon landing, such that larger parachutes may then need to be selected (to satisfy descent rate requirements) which will further increase weight and change your stability margin.

During simulations, it is good to select a ‘nominal’ motor, and a ‘backup’ motor. The nominal motor is used with the stock or modified kit model, and simulations satisfy all requirements. As the build progresses, the backup motor would be used if the weight started to increase beyond what the nominal motor simulations expected and reaching minimum altitude becomes an issue. It is best, however, to try to ‘shave’ weight by any means necessary (or reduce the drag, by polishing to surface somehow).

Don’t forget to verify and understand all the requirements!

12f. Design & Acquisition Notes

13. Rocket Build

13a. Pre-Fit Check

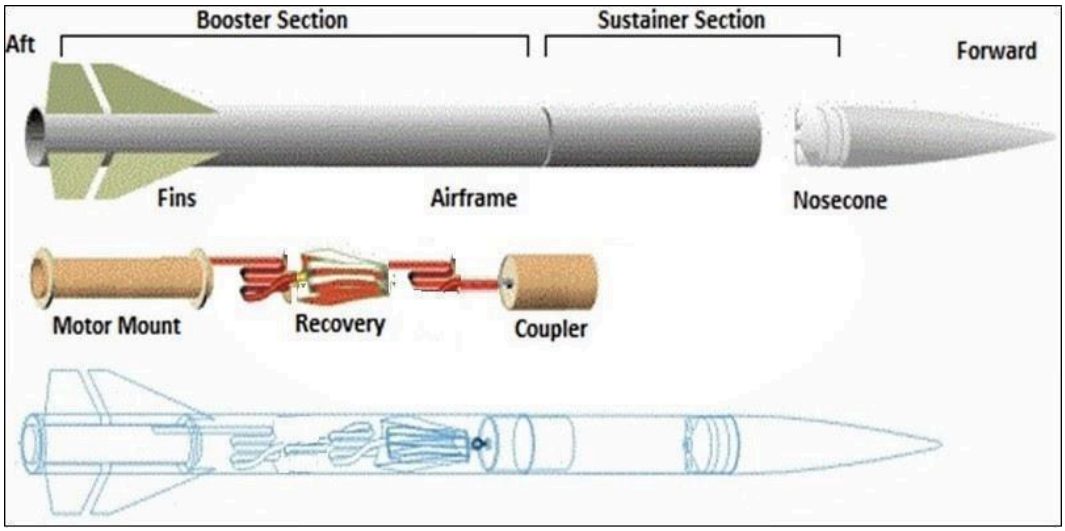


Figure 13-1: High-Powered Rocket Diagram

The key to proper fit and adhesion is to **DRY-FIT** all rocket parts **BEFORE** applying any epoxy to parts. Do not over-sand your parts, a snug fit is required.

Parts to dry-fit (DETAILED ON FOLLOWING PAGES):

- Centering Rings
- Coupler and Bulkplate
- Motor Mount Tube (MMT)
- Airframe
- Fins
- Nose cone

Notes:

i. Centering Ring

Reference: *5e. Centering Rings / Bulkplates*

1. Identify and label centering rings
    a. Mark the centering rings: Forward, Middle, AFT.
2. Rough sand entire length of motor mount tube (MMT) (for adhesion) just enough to remove glassine
3. Sand inside of centering rings so they fit into MMT

13-2: Aligned Centering Rings (top – Forward, middle – Center, bottom – Aft)

4. Install <u>small</u> eyebolt into the **<u>Forward</u>** centering ring
    a. One nut and <u>small</u> washer on the eyebolt side of the centering ring (depends on hardware)
    b. <u>Small</u> washer and nut on the back end of the centering ring
    c. Motor retention will use #8 wood screws to hold the motor retainer to the Aft Centering ring

Figure 13-3: Forward Centering Ring with Eyebolt Installed

<u>Notes:</u>

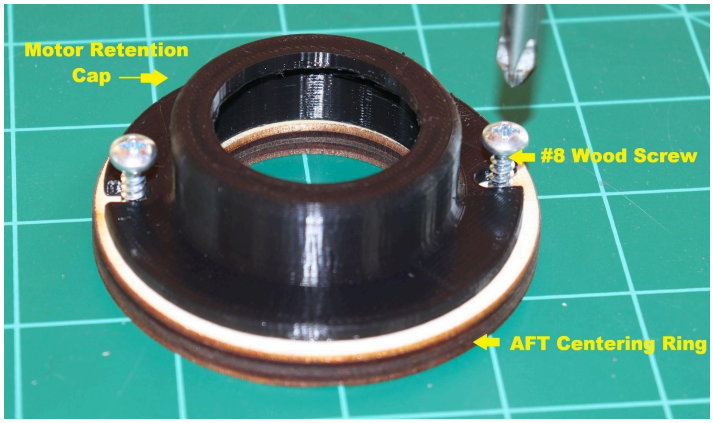


Figure 13-4: Aft ring with Motor retainer with wood screws

ii. Bulkplate

Reference: *5e. Centering Rings / Bulkplates*

1. Install an eyebolt into the bulkplate
    a. One large washer and one nut on the eyebolt side of the centering ring
    b. One large washer and nut on the back end of the bulkplate *(If you only have one washer, the washer should be placed on the back side of the bulkplate)*
2. Dry-fit bulkplate into coupler tube (using eyebolt as handle)
3. Sand outside of bulkplate for proper fit
4. Place the **Coupler Bulkplate** 1/4" from end of the coupler

Figure 13-5: Bulkplate and Eyebolt

Figure 13-6: Eyebolt Installed in Bulkplate

Notes:

Figure 13-7: Eyebolt installed in Bulkplate

iii. Centering Ring Alignment

Reference: *5e. Centering Rings / Bulkplates, 5f. Motor Mount Tube*

1. Centering rings should fit securely but removable on motor mount tube
2. Mark the forward end of the motor mount tube 1/8” – 3/16” for the forward centering ring placement
3. Measure 4-3/4” from the AFT end of the motor mount tube to the AFT end of the Mid centering ring. Place a reference mark
4. Measure 4-3/16 from the AFT mid centering ring to the forward AFT centering ring. Please refer to diagram below.

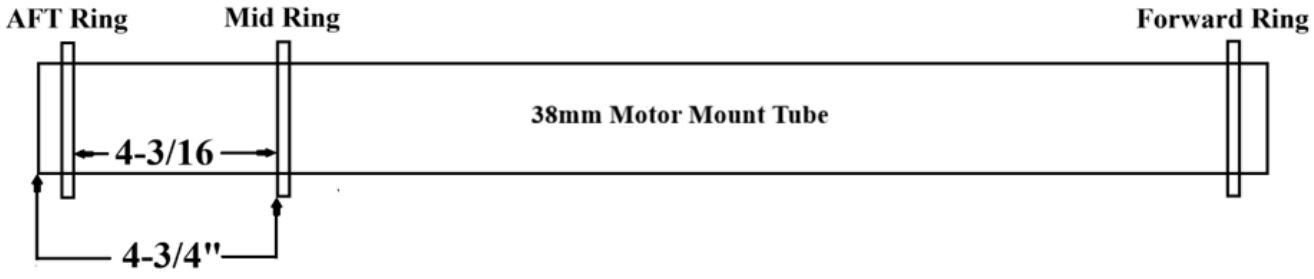


Figure 13-8: Centering Rings and measurements on Motor Mount Tube

5. Before removing centering rings for the next procedure make sure reference marks have been made on the motor mount tube.

Notes:

Figure 13-8: Centering Rings and me

Figure 13-9: Motor Mount Tube with centering rings and centerline

iv. Airframe

Reference: *5c. Airframe*

1. Rough sand around fin slots of airframe tube (for adhesion)

Figure 13-10: Fin Slot

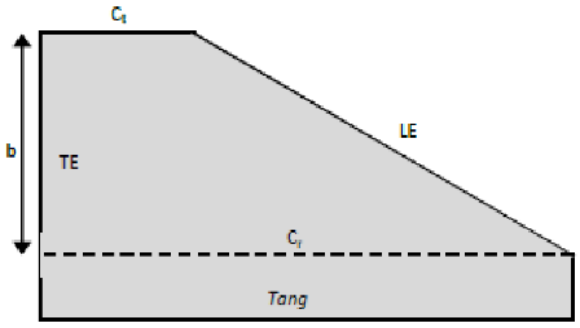


Figure 13-11: Fin diagram displaying tang (tab) that inserts through the airframe and touches the motor mount tube

Notes:

13b. Assembly

i. Epoxy Overview

1. Normally a two-part chemical mixture
    a. A resin
    b. A hardener
    c. Usually mixed in a 1:1 ratio
2. Various Types Based on Material and Strength
    a. Time, Strength
    b. JB Weld – metallic bonding
    c. Silica additive can be used for fin fillets
3. Epoxy Fillets
    a. Creates a strong bond between two surfaces
    b. Applied with a craft stick in single smooth line to create a valley between two surfaces
4. Safety
    a. Always wear gloves prior to mixing and applying epoxy
    b. Work quickly and with small amounts of epoxy at each step

Learn more about epoxy:
https://www.apogeerockets.com/education/downloads/Newsletter186.pdf

Or see the NASA Handbook for construction tips.

Figure 13-12: Epoxy Resin and Hardener

Figure 13-13: Epoxy Application

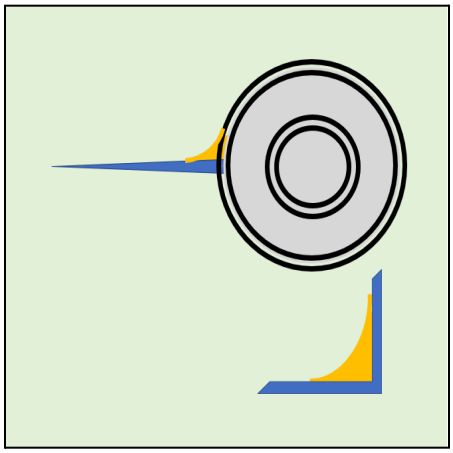


Figure 13-14: Epoxy Fillets

Notes:

ii. Motor Mount Assembly

Reference: *5e. Centering Rings / Bulkplates, 5f. Motor Mount Tube*

1. Remove dry fit centering rings from the motor mount tube.
    a. Mark a centerline down the complete motor mount tube using a straight edge. A door stop molding works well for this. (Figure 13-15)
    b. Align forward centering run ¼" below its reference mark on the motor mount tube. Do the same with the mid centering ring by positioning the mid centering ring ¼" below its reference mark.
    c. Put gloves on and mix a small amount of epoxy, enough to spread around the radius of the motor mount tube at the reference marks: forward and mid centering ring locations (Figure 13-16a)
    d. Slide both forward and mid centering rings up into the epoxy and stopping at the reference marks made earlier. Ensure that a mid centering ring notch is in alignment with the motor mount tube centerline. (Figure 13-16b)
2. Set motor mount assembly aside, allow epoxy to dry

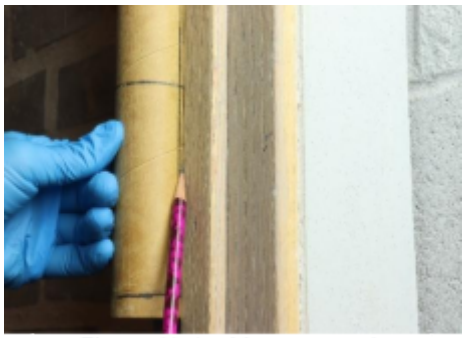


Figure 13-15: Motor mount in doorframe to get tube centerline

Figure 13-16a: Forward centering ring with epoxy before repositioning

Notes:

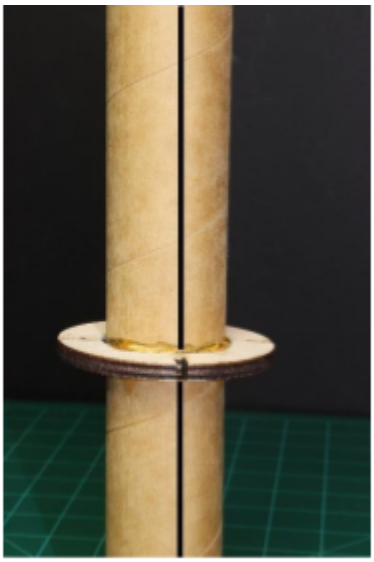


Figure 13-16b: Middle centering ring with notch aligned with motor tube centerline

iii. Coupler Assembly

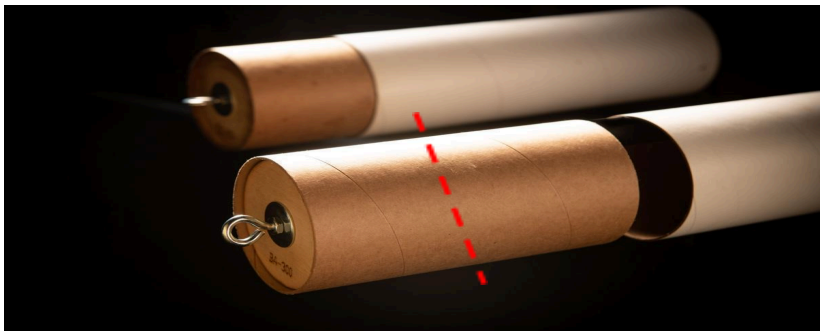
Reference: *5d. Coupler, 5e. Centering Rings / Bulkplates*

1. Dry-fit Step: Fit bulkplate inside coupler, ensuring level fit ~1/4" from the edge
2. Remove bulkplate after ensuring fit
3. Put on gloves before preparing epoxy
4. Prepare small amount of epoxy
5. Dab the threads and nut of eyebolt with epoxy
6. Apply epoxy around inside coupler ~ ¼" from the end
7. Twist the bulkplate into the coupler leaving ¼" gap from the edge of the coupler
8. Apply fillet on forward side of bulkplate
9. Set coupler assembly aside, allow epoxy to dry

Notes:

Figure 13-17: Example of test fit of bulkplate in coupler (Top); Epoxy Application (Middle); Coupler drying (Bottom)

iv. Coupler/Sustainer



Reference: *5a. Parts and Subassemblies, 5d. Coupler*

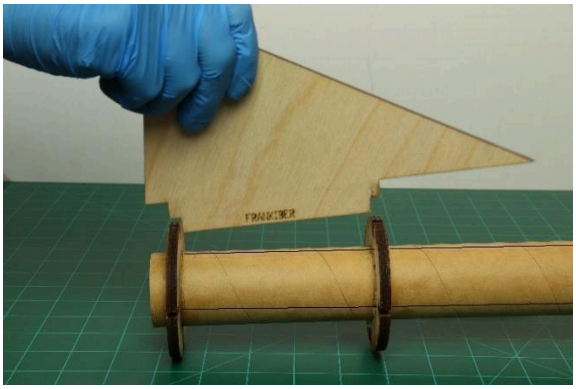
1. Dry-fit coupler to upper airframe
2. Draw a line/mark around the center of the coupler
3. Put on gloves before preparing epoxy
4. Prepare small amount of epoxy
5. Apply epoxy around inside of the sustainer (about 1” from edge)
6. Place coupler into sustainer section aligning aft section of the sustainer with the center line drawn on the coupler
7. Use a twisting motion, to evenly distribute the epoxy
8. Set aside, let epoxy dry

Figure 13-18: Example of Where to Draw Line on Coupler (Top); Epoxy Application (Middle); Coupler Placement (Bottom)

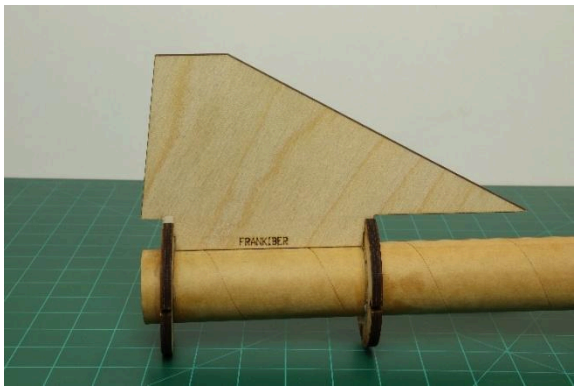
Notes:

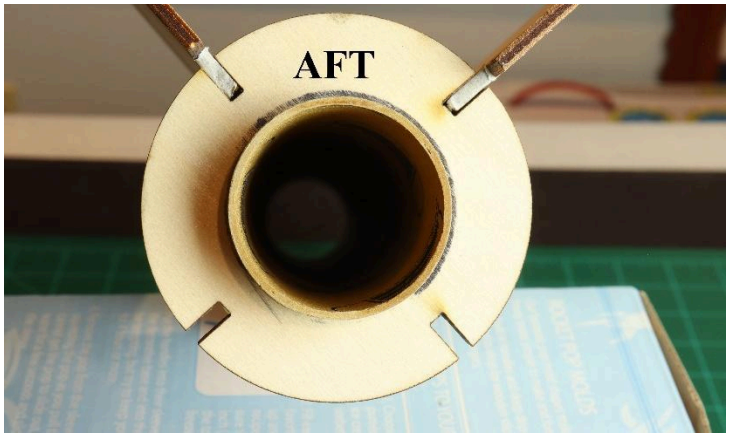
v. Fin Installation

Ensure that middle centering ring epoxy has set before working these steps.



1. Install Aft Centering ring at reference mark.
    a. Test fit fins into the LOC-in fin slots. Sanding may be required. Install all four fins.
2. Once all fins fit securely, put on gloves to mix epoxy.
    a. Carefully remove fins.
    b. Apply epoxy to the Tang (tab) of the fin and install fin into the LOC-in slots.
3. Make sure the Aft Centering ring is also engaged fully with the LOC-in slots and the fin Tang (tab) is making full contact with the motor mount tube.
    a. Let epoxy have time to begin to cure (5 minutes).
4. Repeat procedure until all fins are epoxied into place.
5. Allow epoxy several hours to cure before manipulating motor mount assembly.





vi. Nose Cone

Reference: *5b. Nose Cone*

1. Fit the nose cone to the sustainer
    a. Should not require sanding
2. Drill a single hole at location shown, through both airframe and nose cone with awl or 1/8” drill bit
3. The nose cone is then attached to the airframe with a sheet metal screw (longer pointed screw) to keep the nose cone from separating in flight (but allow for removal of nose cone)
4. Seam along nose cone (flash) may be sanded with fine grain sandpaper (while waiting for fin epoxy to set)

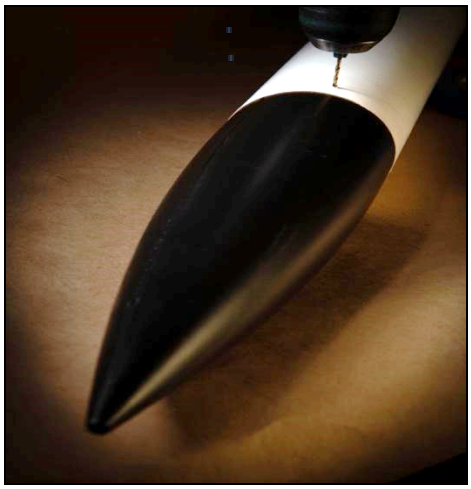


Figure 13-23: Drilling the Airframe and Nose Cone

Notes:

Figure 13-24: Red Dot indicates mark for drilling through airframe and nosecone

vii. Rail Button Alignment

Reference: *5h. Rail Buttons*

Rail Guide System

- 1010 rail guide (it's a 1.0" x 1.0" rail)
- 6' (or 72 inches) in length

Rail Button Measurement

1. Temporarily position a 4"- 5" length of tape between the fin slots. Place a mark on tape, just inside of both fin slots.
2. Remove the masking tape
3. Fold masking tape in half on non-stick side, matching the two fin slot reference marks
4. Make a crease at the middle location
5. Place the strip of masking tape back on the airframe matching lines to the fin slots
6. Mark the center point at the crease line, making a line halfway between the two fin slots on the aft portion of the booster section
7. Place the line marked on the airframe parallel to a door jam
8. Draw a line halfway up the airframe, creating the rail button centerline
9. This is the line the rail buttons will be installed on after the fins have been installed

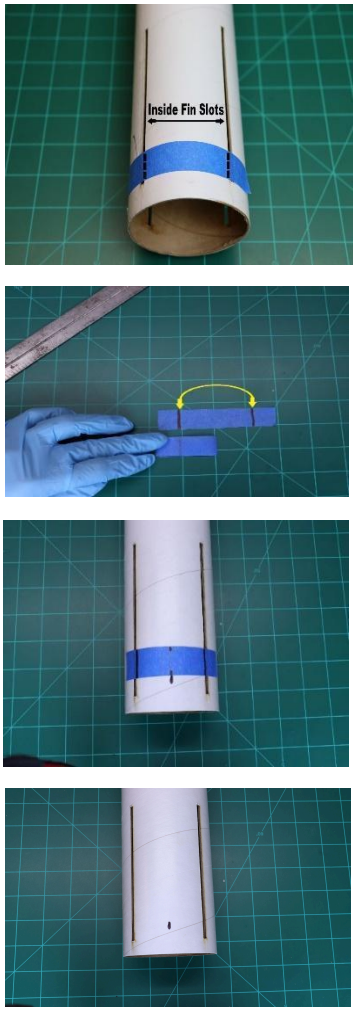


Figure 13-25: Measuring Fin Slots (Top); Measuring Half Way between Two Fin Slots (Top Center); Drawing a Line at the Half Way Mark on Tape (Center); Drawing the Half Way Mark on the Air Frame (Bottom Center); Markers to Indicate Rail Button Placement/Alignment (Bottom)

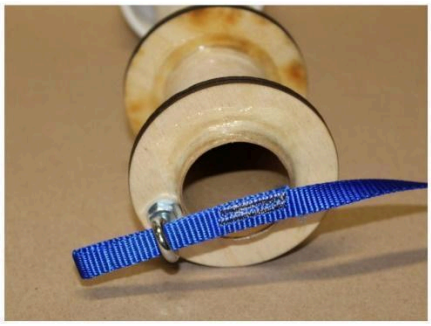
Figure 13-26: Rail Buttons are Drilled into the Mid-Centering Ring (Top) and Aft-Centering Ring (Bottom)

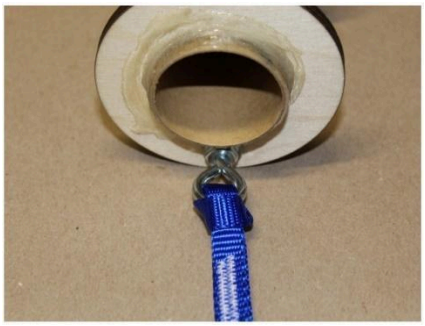
viii. Motor Mount

Reference: *5f. Motor Mount Tube, 8. Recovery Hardware*

Figure 13-19: Shock Cord and Quicklink

1. Attach the shock cord to the forward centering ring eyebolt using a quicklink in the shock cord loop and the eyebolt
    a. A double slip square knot can be used if a loop is not on the shock cord
        i. Attach shock cord by feeding the cord through the eyebolt
        ii. Feed the opposite end of the shock cord through the sewn loop
        iii. Pull shock cord tight





2. Re-coil shock cord in ~8 - 10” in length, securing with masking tape when done

3. Leave enough length on the eyebolt end of the shock cord such that it will come out the end of the forward opening on the airframe

Figure 13-20: Shock Cord Attached to Eyebolt

4. Stuff the bundle of cord inside the motor mount tube

ix. Fin Can Installation

Reference: *5i. Fins*

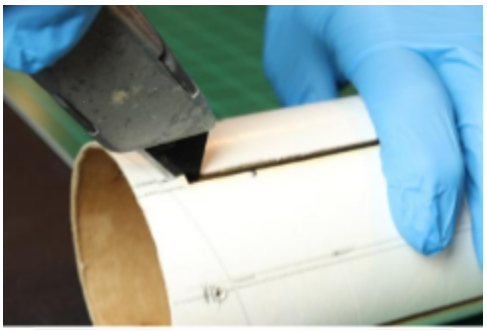


Figure 13-29: Fin slot lines being cut out from airframe

Dry-fit Step:

1. Draw straight lines the thickness of the fin slots at the Aft end of the rocket.
    a. Use a hobby/utility knife or scissors to cut the airframe
2. Position airframe vertically, Aft end up (fin slots up)
3. Test fit motor fin can into airframe – use paper shims to assist with insertion
4. Sand outer edge of centering rings as needed
    a. Avoid making burrs on the inner edge of the airframe
    b. If a burr occurs, gently sand the inner edge of the airframe
5. Remove MMT after ensuring proper fit

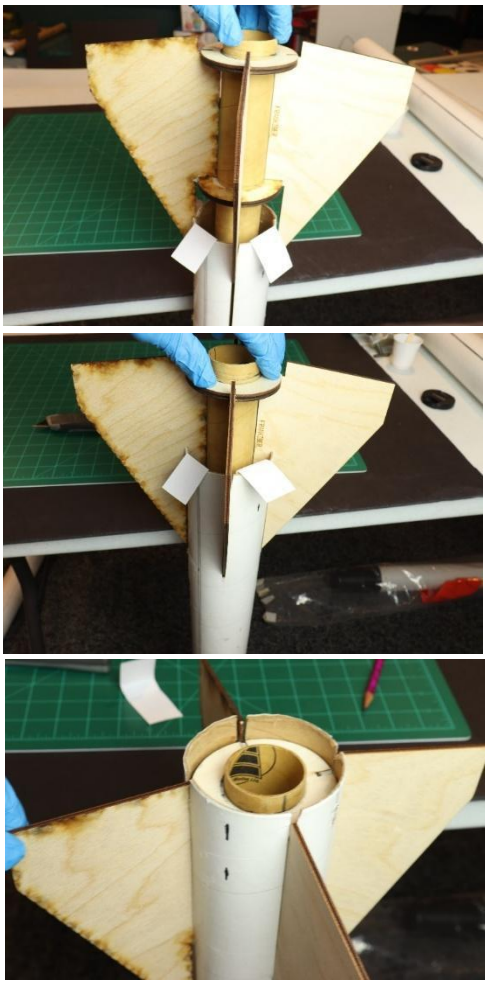


Figure 13-30: Example of paper shims helping slide in fin can assembly and final fin can install goal

Installation:

1. Put on gloves before preparing epoxy
2. Prepare small amount of epoxy
3. Epoxy inside parameter of the airframe just in front of the forward center ring location using craft stick taped onto dowel rod
4. Line up fin can with fin slots and gently fit fin can into airframe (utilize paper shims) until Aft Centering ring is about ½" recessed into the airframe or the forward fin tang (tabs) are flush against the forward edge of the fin slots
5. Apply tape over motor retainer screw holes.
6. If needed, prepare a small amount of epoxy
7. Apply small fillet of epoxy around edge of aft centering ring and airframe being careful around motor retainer screw holes

Notes:

x. Outer Fin Fillets

Outer Fin Fillets (Prep)

1. Fillet tape must be installed parallel to fin, along the full length about 1/4" away from corner
    a. On both fin and tube (red arrows)
2. Repeat fillet tape for all four fins
    a. Will be 16 strips of tape for entire rocket for fillets
3. With the rocket positioned as shown, place 4 pieces of tape to create 4 dams to fill in the small gap between the fin and tube, on top of fillet tape
4. Place tape under fin long enough to cover gap
    a. Must be tight in corner, so epoxy does not leak through (green arrows)

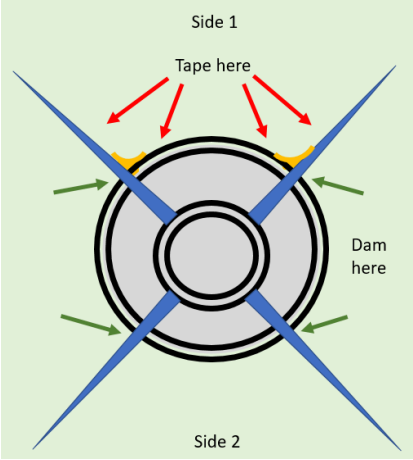


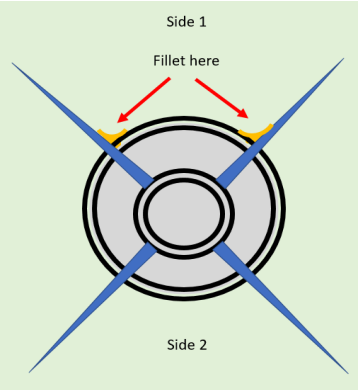
Figure 13-34: Tape and Dam Diagram

Figure 13-35 (Left); Tape Locations for Epoxy Fillets (Center); Dam Locations Right)

Notes:

Outer Fin Fillets (Epoxy)

1. For Side 1 (as shown):
    a. Put on gloves before preparing epoxy
    b. Prepare small amount of epoxy
    c. Fill in gap with epoxy (both fins) – let set few minutes, ensure epoxy is not leaking past dam on underside
    d. Epoxy along entire length of fin root, create smooth fillets (both fins)
    e. Remove dam tape from underside after 10 minutes
    f. Remove fillet tape from Side 1 (tube and fins)
    g. Let set up for 10 – 15 minutes
2. Rotate rocket 180 degrees, and repeat for Side 2
3. Sides 3 and 4 do not require a dam step, fillet only



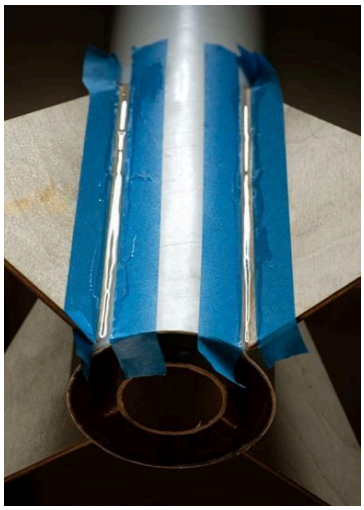


Figure 13-36: Fillet Diagram (Top); Fin Fillet Fill (Bottom)

Notes:

xi. Vent Holes

Reference: *7b. Center of Pressure*

- Vent (pressure relief) holes allow the pressure inside the rocket to equalize to the external atmosphere.
- Usually 1/8”-1/4” in diameter

1. Drill first hole with an awl or 1/8” drill bit in the middle region of the sustainer
2. Drill second hole an awl or 1/8” drill bit in the booster region above the motor mount tube/fins

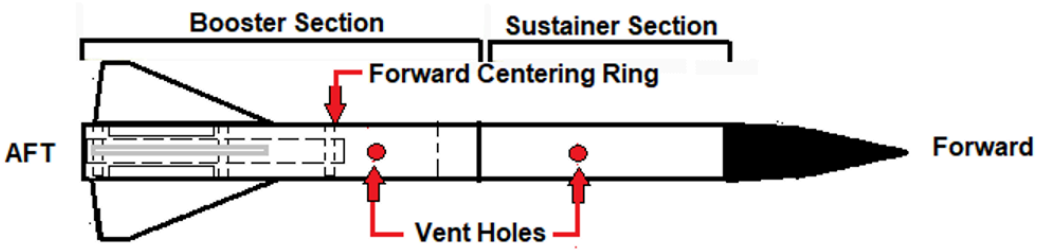


Figure 13-27: Vent Holes Indicated in Red on Rocket Airframe

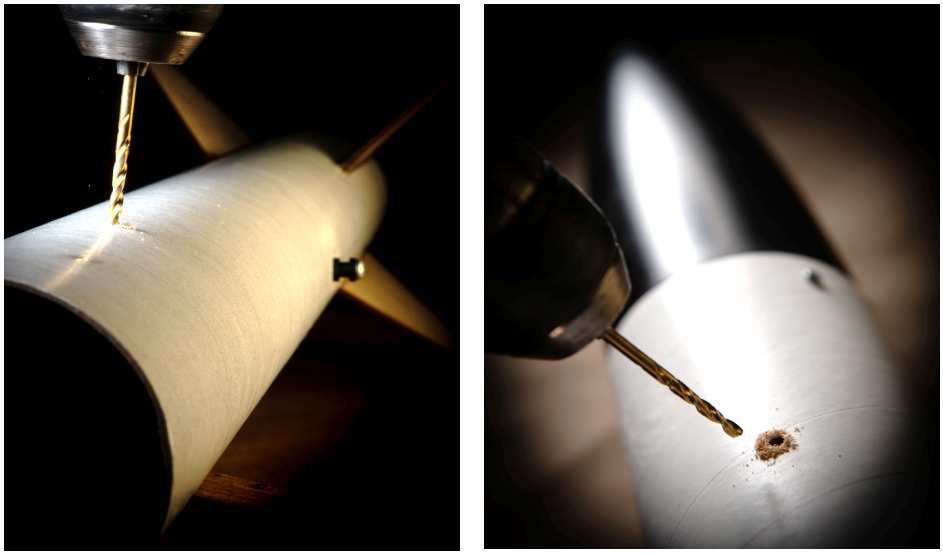


Figure 13-28: Drilling Vent Hole in Booster Section to Not Interfere with Rail Buttons Alignment (Left); Drilling Vent Hole in Sustainer Section (Right)

Notes:

xii. Rail Button Installation

Reference: *5h. Rail Buttons*

Rail Buttons are installed so that your rocket can be fit to a standard launch rail

The bottom rail button will be installed into the aft centering ring, while the top rail button will be installed into the airframe (with backing nut – or into forward centering ring)

1. Mark an 'X' on the vertical line where it intersects the aft CR
2. Mark an 'X' on the vertical line near the forward CR
    a. Attach dowel to your ruler, insert into forward end of airframe until it touches the forward CR
    b. Mark line on dowel indicating end of airframe, remove and place on top of airframe to use as "measuring tape"
3. Drill holes using a 1/8" drill bit – ensure hole is perpendicular to surface
4. Dab epoxy in hole
5. Attach rail buttons and pointed screws into place

Figure 13-38: Rail Button Alignment on Launch Rail (Top); Marking an 'X' on the Vertical Line Where the Line Intersects with the Aft and Forward Centering Ring (Bottom)

Notes:

xiii. Recovery Installation

Reference: *8. Recovery Hardware*

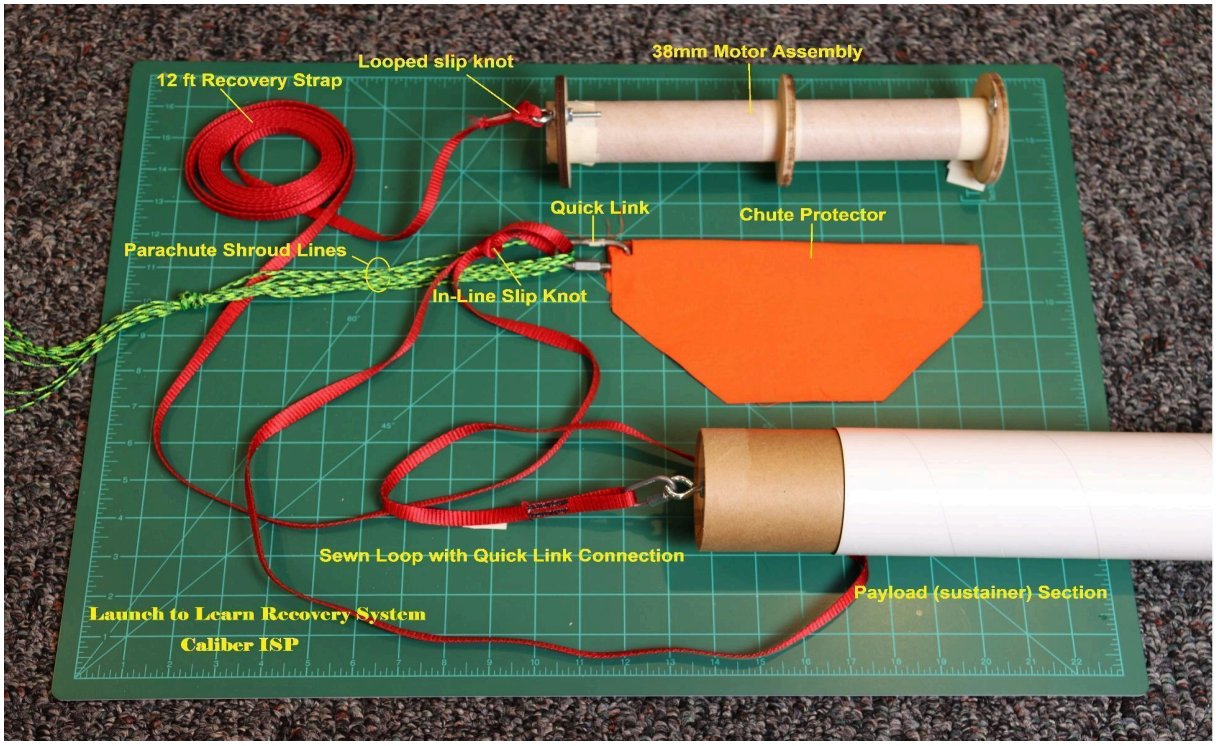


Figure 13-39: Recovery Supplies

Failed parachute deployment: https://www.youtube.com/watch?v=M2G-UVCO7u0

1. Push a shock cord out through forward end of airframe, remove masking tape
2. Tie an in-line slip knot in the shock cord (1/3 point of the total shock cord length)
3. Untangle parachute shroud lines
4. Place arm through the shroud lines

Figure 13-40: Masking Tape and Shock Cord (Left); Knot Location on Shock Cord (Right)

5. Attach the parachute to the parachute with a looped slip knot
    a. Feed shroud lines through sewn loop in shock cord end
    b. Pass parachute through shroud lines
    c. Pull parachute tight

Figure 13-41: Overview Knot Step 1

Notes:

Figure 13-42: Overview Knot Step 2

Figure 13-43: Overview Knot Step 3

xiv. Parachute Preparation

1. Make sure the parachute shroud lines are untangled and then lay it flat on the table.

Figure 13-44: Step 1 of Parachute Prep

2. Position one shroud line nearest you, then start folding the parachute gores into angular sections by bringing each consecutive shroud line over the first and evening out the fold up to the tip.

Figure 13-45: Step 2 of Parachute Prep

3. Once all goers are folded with all shroud lines together, it should appear like this.

Figure 13-46: Step 3 of Parachute Prep

4. Fold the parachute like a zigzag.

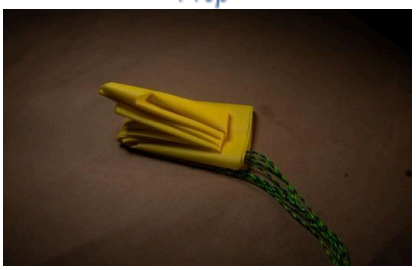


Figure 13-47: Step 4 of Parachute Prep

5. Tightly roll the parachute into a cylinder. Ensure all shroud likes are still untangled and together.

Figure 13-48: Step 5 of Parachute Prep

6. Neatly bring the shroud lines together in a zigzag fashion and place them next to the parachute at one corner of the parachute protector or Nomex cloth.

   a. Watch demo of alternate method of small Nomex cloth

Figure 13-49 Step 6 of Parachute Prep

7. Start rolling the Nomex over the parachute and shroud lines. Keep it snug. Then place the shock cord onto the Nomex using the same method as the shroud lines.

Figure 13-50: Step 7 of Parachute Prep

8. Fold the left and right corners of the Nomex inward, then continue rolling the bundle tightly, keeping all of the shroud line inside the Nomex cloth.

Figure 13-51: Step 8 of Parachute Prep

9. Place the newly wrapped parachute bundle into the booster section of the rocket

14. Certification Launch Preparation

14a. Safety Overview

Launch operations require TRA or NAR personnel in order to complete a launch safely. At a minimum, the personnel required are:

1. Range Safety Officer (RSO)
    a. The goal of the RSO is to minimize the risks to personnel and property involved in the handling, preparation, and launch operations of model and high-powered rocket launches
2. Launchpad Safety Officer (LSO) / Launch Control Officer (LCO)
    a. The LSO/LCO is responsible for determining the status of range operations: site, airspace, and weather.
3. Launchpad Manager
    a. The Launch Pad Manager will assist you with the launch prep
4. Rocketry Mentor
    a. TRA/NAR certified at a level equal to or above the motor you will use to certify
    b. May guide, direct, and assist you as you prepare your rocket and go to the launch pad

14b. Launch Site Materials and Supplies

Materials and supplies that may be helpful at the launch site:

- Drill
- Screwdriver
- Dog Barf
- Epoxy
- Masking Tape
- Screw
- Washer
- Nut
- Etc.

Figure 14:1 Launch Site Preparation Area

14c. Pre-Launch Rocket Preparation

Prior to the certification launch, you will need to prepare the motor and the parachute. Once you are assigned a launch pad, you will need to place your rocket on the launch rail and prepare the rocket for launch.

- Motor
- Recovery
- Launch Pad

14d. Motor Preparation

For a certification launch, you will need to prepare your own motor under supervision of the club RSO (or your mentor).

Figure 14-2: Certification Motor and Packaging

1. Obtain your motor from the RSO
   a. WSGC provides one certification motor
2. Perform your own motor prep
   a. Remove certification motor from protective cardboard tube
   b. Delay Adjustment
   c. Black Powder
   d. Motor Installation
3. RSO will observe your motor prep
   a. It is not a test, it is a hands-on learning experience
   b. If you don't understand something, ask questions
4. Insert your motor in rocket and install motor retainer
5. Suggest you tape the ignitor to the outside of the rocket for later

i. Remove from Container

Your certification motor will come in a protective cardboard tube when shipped. Basic motor specs are found on the label. You will remove the motor from the tube – it may also be sealed in a plastic bag, so remove that as well.

The motor will have parts contained within:

1. Motor
2. Ignitor Leads
3. Pyrogen
4. Vial of Black Powder
5. Red Forward Plug
6. Baffle Washer

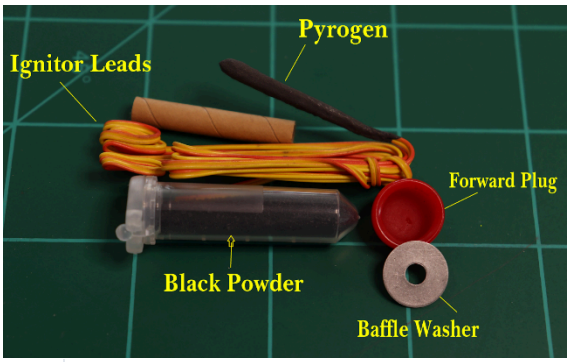


Figure 14-3: Certification Motor Parts

Ensure the motor is intact and there is no obvious damage from shipping. You will note the nozzle end and the forward charge well end (see figure 6-3 in Commercial Motor section on page 25). You will need to adjust the motor ejection charge delay, using an Aerotech ejection delay tool.

Figure 14-4: Delay Tool

ii. Delay Adjustment

You will need to determine the expected time to apogee. The motor should come with a standard 14-second delay. You will need to remove the amount of time (in seconds) from the standard delay, for your rocket flight.

Figure 14-5 Removing seconds from delay

For example, you simulate your flight to reach apogee in 12 seconds. You would then remove 2 seconds from the standard delay. The tool will have various adjustments; you will find the one that removes 2 seconds. This will ensure the parachute deploys at the proper time.

1. Adjust the motor ejection charge delay
   a. Use the delay adjust tool to remove seconds from delay
      i. Adjust the tool for proper amount
      ii. Screw the tool into the end
   b. Shake flakes out – discard
   c. Install Baffle Washer

Figure 14-6: Install Baffle Washer

iii. Black Powder

Once the delay is adjusted properly, you will remove the shavings. You will then pour the black powder from the vial into the end, and cover with the red rubber cap.

   d. Pour black powder into deployment well
   e. Insert Plug Cap on end

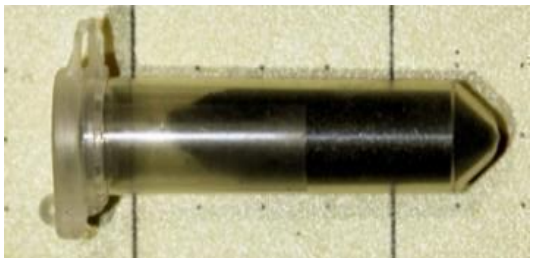


Figure 14-7: Black Powder Vial

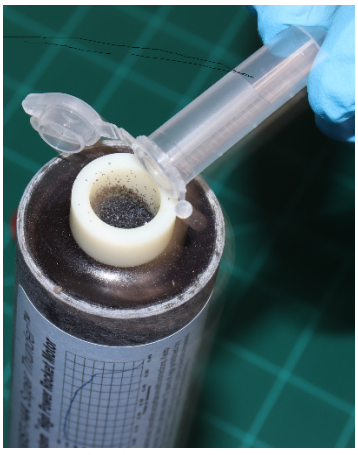


Figure 14-8: Pour Black Powder into Deployment Well

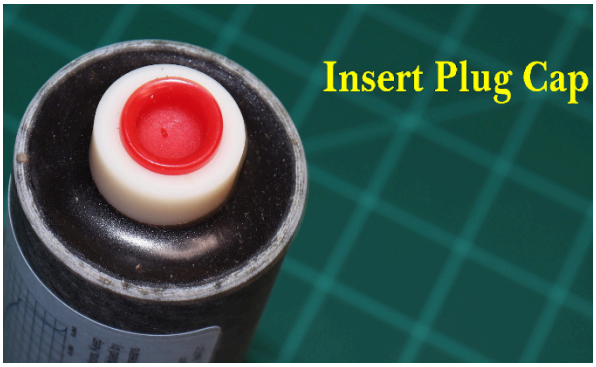


Figure 14-9: Pour Black Powder into Deployment Well

iv. Motor Installation

Finally, you can insert your motor into your rocket. Ensure that it inserts smoothly. DO NOT apply too much force. If the motor cannot be installed smoothly, check for epoxy inside the motor mount tube. If there is epoxy, you may need to clean the tube.

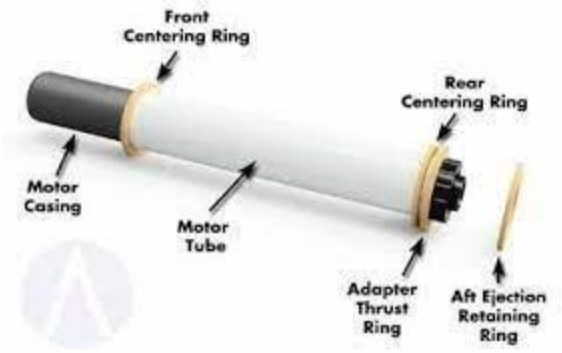


Figure 14-10: Motor Installation

Once the motor is installed, ensure it is flush with the aft motor mount tube (all the way in). You can then install your motor retainer.

We recommend that you tape the motor ignitor (keep in bag), to the exterior of your rocket. This ensures that you do not misplace the ignitor before you reach the launch pad. You will also need the tape once you get to the launch pad.

Fill out your launch card with the motor details.

14e. Recovery Preparation

i. Parachute

1. Know how to fold / install your parachute into your rocket (as previous demo)
   a. Ensure parachute and protectors are **attached** inside of quick-links
   a. Ensure your quick-links are attached to rocket and **closed**
   b. Ensure to use parachute protectors and/or 'dog barf' to protect your nylon parachute from the hot ejection charge gases
      . Ensure parachute protection is on bottom (where ejection charge is) and parachute is on top
2. If your certification flight occurs in adverse conditions
   a. Consider using a smaller parachute than the stock parachute provided, to prevent drift
      i. Your Rocketry Mentor may be able to assist you
   b. You may 'reef' your shroud lines using tape (or by tying a knot part way up the shroud lines)
      i. This will decrease the parachute open diameter
      ii. This will increase descent rate (decrease drift)
   c. Ensure your mentor or RSO is aware of any adjustments you have made prior to flight

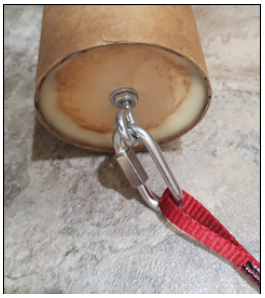


Figure 14.11: Parachute and Protectors Attached Inside of Quicklinks (Top); Quicklink Attached to Rocket and Closed (Bottom)

Figure 14-12: "Reef" the Shroud Lines Using Tape

3. At Tripoli Wisconsin (only) - additional recovery techniques are used for certification flights (when flying in adverse conditions)

ii. Jolly Logic

Although unlikely, in the case of high wind conditions on certification day, you will be using a Jolly Logic parachute release (for Tripoli Wisconsin certifications).

The Jolly Logic device will prevent your parachute from opening at motor ejection. The parachute still deploys (exits the rocket), but the canopy does not open. The Jolly Logic can be used to 'release' the canopy at a lower altitude. This prevents unwanted drift in high wind conditions, allowing you to recover it sooner and eliminate the chance of losing it.

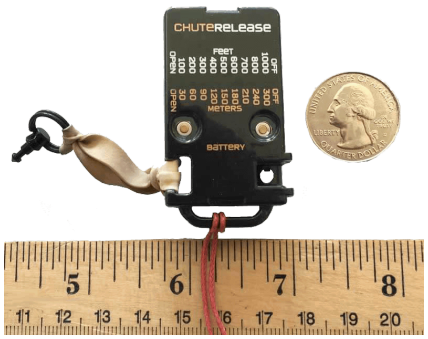


Figure 14-13: Jolly Logic Parachute Release

https://jollylogic.com/products/chuterelease/

a. Jolly Logic Parachute Release
   i. This device will be loaned to flyers
   ii. This device will not open the parachute until a much lower altitude
      1. AKA streamer (or tumble) recovery from apogee

iii. RF Tracker and Antenna

For Tripoli Wisconsin certification flights, we may also have RF tracking devices available for use. This small device goes inside your rocket (or simply tape to your shock cord). The device emits a radio signal (transmitter) on a specific frequency that a hand-held device, sometimes called a Yagi antenna (receiver) can track. The transmitter has a range of one mile.

You will need to remember the frequency of the transmitter (which is on the inside cap of the device). The antenna needs to be tuned to this frequency. The antenna will emit an audible pulse that is louder in the direction of the transmitter, and softer if pointed away from the transmitter.

Figure 14-14: RF Tracking Device (Left); Yagi Antenna (Right)

https://www.com-spec.com/rcplane/index.html

a. RF Tracker and Antenna
   i. This device will be loaned to flyers
   ii. This device will emit an RF signal that allows you to track on ground
      1. It can be flown in your sustainer section or taped to shock cord

These devices ARE NOT REQUIRED for certification flight

a. If you plan to use a device, be familiar with its operation prior to flight

a. Ensure your mentor or RSO is aware of any devices used in flight

14f. Launch Pad Preparation

Once you have prepared your rocket for the certification launch and have completed the pre-launch checklist check in with the LCO for your pad assignment. Turn in the launch card and proceed to the Launch pad when the "Range is Open." The Launch Pad Manager (or Mentor) will assist you with the launch prep.

i. Steps to Prepare Launch Pad

1. The rail will be tilted over for loading
    a. Slide your rocket on the rail, aligning your rail buttons in the grooves in the rail
2. Your rocket should be on top – your rail buttons down
    a. Slide all the way down until it hits the stops
3. Remove the ignitor from the bag
    a. Pull the exposed ends apart a few inches
    b. Strip an inch of wire covering from the exposed ends
4. Insert the bulb end of the ignitor into the motor
    a. Push the bulb as far as it can go, until you feel it stop
    b. Tape the kinked end to the bottom
    c. Ensure that the wire is secure and will not fall out
5. Grab the alligator clips (on the ground)
    a. Wrap each ignitor lead around each clip
    b. Ensure the leads do not contact each other or the rail

Figure 14-15: Ignitor (Top); Alligator Clip (Bottom)

14g. Rocket Retrieval

There are several things you should be aware of prior to your rocket retrieval (for certification at Tripoli Wisconsin).

i. Landmarks / Hazards

It is always a good idea to scout out the area (at least look at a map) to see where your rocket may land, and the best way to get there. There are a few roads in the launch area that you may drive on for better access to where your rocket may have landed.

In good conditions, certification flights at Tripoli Wisconsin usually land within 2500 feet radius of the launch pads (the green circle). In less-than-ideal conditions, certification flights have landed up to 4000 feet from the launch pads (the yellow circle).

You should also try to note the hazards in the area, which your rocket may land in or nearby. Be aware of the water hazards to the north. There are also numerous tall trees and tall grass in the area. The ground can be soft and muddy as well, so be prepared.

ii. Range and Line of Sight

If you are lucky, in real calm conditions, your rocket may land within a few hundred feet of the launch pad. This area contains some tall grass but is otherwise free of hazards. You will simply be able to walk

out and collect your rocket – your parachute is brightly colored to assist you with spotting it from a distance on the ground.

When your rocket is descending, you should try to ascertain a range and a line of sight. You can try to use the tree lines to determine the range. You should also use other objects to determine a line of sight. As you get closer to the area where you suspect your rocket has landed, you can use the objects to look back to the launch site, to get your bearing.

Figure 14-16: Richard Bong State Recreation Area Aerial Map

Next, you would need to determine if you can simply walk directly towards your rocket, or if you should drive around to another road / access point and walk from there. We also suggest you go in pairs to recover, assisting each other.

For your own safety, please do not enter the water to recover your rocket. Please do not attempt to climb trees either. Tripoli Wisconsin has poles that you can borrow to help recover your rocket. Worst case, it may be unrecoverable even though it is in plain sight.

iii. Post-Flight Inspection

You must return your rocket to the RSO for inspection following your flight. Your rocket must be free of any damage for your certification to be complete. If your rocket is unrecoverable, notify the RSO.

15. Certification Launch Overview

Level 1 certification allows flyers to fly high-powered rockets with a total installed impulse up to 640 newton-seconds. WSGC simulates the LOC Launch 2 Learn kit with an Aerotech H100W DMS.

15a. Certification Launch Requirements

i. Steps for Completing Your Launch:

1. When arriving at the launch site, you are expected to abide by the presiding organization's safety procedures
2. Requirements at the certification launch
    a. Review pre-launch checklist
        i. Complete launch card and submit to TRA/NAR
        ii. After rocket recovery, return to HPR Launch Operations area
        iii. Properly dispose of motor
        iv. Sign and submit TRA Membership Application to Frank Nobile (at Wisconsin Certification Launch only)
        v. Pack rocket for return home
3. Fees associated with certification launch
    a. Certification launches will take place during launch competition weekend and costs of the motor and launch fee are included with your acceptance and participation in the workshop.

15b. Certification Rocket Overview

Tripoli Requirements

- Register
- Airframe
- Recovery
- Motor
- Electronics – Not required
- Certification Flight
- Post flight inspection

i. Airframe

The rocket must be built by the flyer. The rocket shall have a display on the exterior identifying the calculated center of pressure. The rocket must be of "conventional rocket design". "Odd Rockets" including flying pyramids, saucers and flying spools will not be allowed for any certification flight. The rocket may be either a kit or scratch built. Scratch built rockets may contain commercially built components.

ii. Recovery

Standard parachute recovery is required. Non-parachute recovery methods (e.g., tumble, helicopter, gliding, etc.) are not permitted for certification flights. If the rocket is using dual deployment, the first recovery event may be via a drogue-less or streamer if the main or second event uses a standard parachute.

iii. Motor

The certification flight must be with a single certified H or I motor (tested total impulse between 160.01 and 640.00 n-sec). Staged and/or clustered rockets may not be used for certification flights. The flyer shall be observed during the assembly (if a reload or hybrid) and preparation of the motor by the certifying member or their designated representative.

iv. Electronics

Electronics are not required for Level 1 certification flights.

15c. Certification Pre-Flight

After completing construction of your Level 1 rocket, the next task is to locate the nearest launch site. Level 1 certification flights may take place at any insured launch that is sponsored by TRA or NAR. It is recommended to call the club launch coordinator beforehand to verify the FAA waiver, field conditions, and if Level 1 certification flights are honored.

The certifying member (i.e., Prefect, TRA Director, or Technical Advisory Panel (TAP) member) must be present and witness the certification flight. The certifying member must witness the rocket ascend in a stable manner and descend in a stabilized manner controlled by the recovery system.

You can find U.S. launch locations at:
http://www.tripoli.org/Prefectures
https://www.nar.org/content.aspx?page_id=225&club_id=114127

i. Pre-Flight Overview

When you arrive at the launch site on launch day, you will need to:

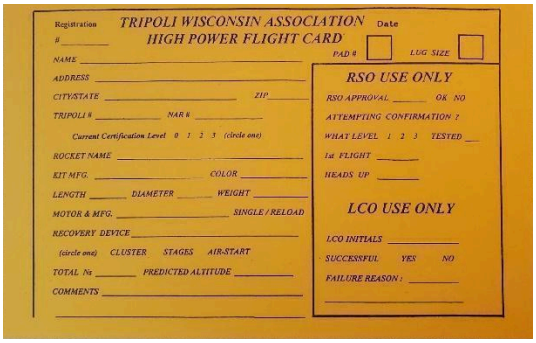

TRIPOLI WISCONSIN ASSOCIATION
HIGH POWER FLIGHT CARD
RSO USE ONLY
LCO USE ONLY

Figure 15-1: Tripoli Wisconsin Launch Card

1. Obtain a launch card from the club organizer table
2. Prepare your rocket for flight
3. Ensure all of your bonds/fillets are intact
   a. Ensure all of your parts fit – no excess epoxy
4. Prepare your parachute as instructed in the workshop
5. Ensure your nose cone is attached
   a. You may fly a 'payload' if you wish
6. Ensure you have a piece of tape with you
7. Prepare your motor (see the RSO/Rocketry Mentor for motors)
8. Ensure you have your motor retainer with you and tools to tighten it
9. Fill out / turn in your launch card to the Launch Control Operator (LCO)
   a. LCO will assign you a launch pad (numbered)
10. Proceed to launch pad when range is open
    a. Usually, a Launch Pad manager (or rocketry mentor) will assist you with launch prep
11. Return to pit area to watch your launch!
12. Observe and recover your rocket
13. Return rocket to Range Safety Officer (RSO) for inspection and sign off

ii. Pre-Flight Inspection

Prepare the following information for the LCO when you turn in your launch card.

1. TRA/NAR Membership #
2. Kit Manufacturer
3. Center of Gravity (CG) and Center of Pressure (CP) – Mark the CG and CP on the rocket
4. Motor Type and Manufacturer
5. Type of Recovery Device
6. Length, Diameter, and Weight of Rocket
7. Predicted Altitude (Remember to conduct a simulation in RockSim)

15d. Certification Post-Flight

The rocket must be presented to the certifying member for inspection. If the rocket cannot be recovered, but can be inspected in place (power lines, tree, etc...), this is acceptable. The certifying member shall inspect the rocket for excessive damage. Excessive damage shall be considered damage to the point that if the flyer were handed another motor, the rocket could not be put on the pad and flown again safely. Damage caused by wind dragging will not cause a disqualification.

i. Certification

Upon a successful flight and retrieval of your rocket, the following steps will take place:

1. The RSO will inspect your certification rocket
2. The RSO will sign your certification form
   a. WSGC will reimburse L2L participants for the first year of TRA or NAR membership
3. WSGC will pay L2L participants TRA or NAR membership for *certification flights* conducted at Richard Bong Recreation Area
4. Tripoli Membership cards will be mailed to participants address provided on form
   a. Membership is valid for one year
   b. Membership must be renewed by participant annually
   c. TRA Certification is valid at NAR launches
   d. Participants may obtain level II Certification; however, WSGC does not cover the cost of certification.

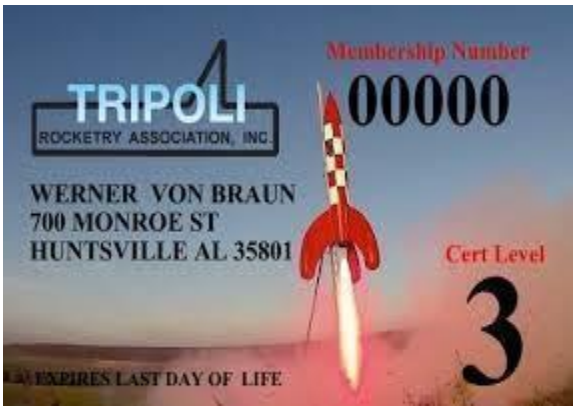


Figure 15-2: Tripoli Membership Card Example

Individuals certifying at a local launch site may certify through the National Association of Rocketry. NAR and TRA certification/membership procedures are different. Please consult your team's mentor for further information.

ii. Non-Certification

Any of the following will result in non-certification for a certification flight:

1. Motor CATO (CATastrOphic failure of a rocket engine)
2. Excessive damage
3. No recovery system deployment or tangled recovery system deployment
4. Rocket drifting outside the specified launch range
5. Components coming down not attached to the recovery system
6. Any other violation of TRA safety code associated with this particular flight
7. Any other legitimate reason the certifying member deems merits non-certification

5e. Rocket Transport

When preparing to bring your certification rocket home, you will need to take the necessary steps to clean your rocket and prepare for the transport.

1. Cleaning and packaging rocket for public transportation
    a. Remove / dispose of the motor
    b. Wipe out any black powder residue
    c. Disassemble any components
2. Transporting rocket from Kenosha, WI
    a. Checked bag for airline flights
    b. Ship from hotel (confirm this option with the hotel)
        i. Boxes, bubble wrap, and labels will be available for shipping for certification launches conducted at Richard Bong Recreation Area
3. Shipping costs may be submitted with Launch 2 Learn travel expenses

15f. Certification Launch Notes

16. Reference Sheet

16a. WSGC Resources

Wisconsin Space Grant Consortium (WSGC)
https://spacegrant.carthage.edu/

WSGC Website Registration Page (Login/Registration)
https://spacegrant.carthage.edu/about/login/

16b. NASA Resources

NASA Space Grant Consortium(s)
https://www.nasa.gov/stem/spacegrant/home/Space_Grant_Consortium_Websites.html

NASA Systems Engineering Webinar Series
https://www.nasa.gov/smallsat-institute/systems-engineering-series/

NASA Space Flight Program and Project Management Handbook
https://ntrs.nasa.gov/citations/20150000400

16c. Collegiate Rocket Launch (CRL) Resources

CRL Website
https://spacegrant.carthage.edu/funding-programs/undergraduate/rocket-competitions/collegiate/

CRL Resources
https://spacegrant.carthage.edu/students/resources/

16d. First Nations Launch (FNL) Resources

FNL Website
https://spacegrant.carthage.edu/first-nations-launch/

FNL Resources
https://spacegrant.carthage.edu/first-nations-launch/resources/

16d. Apogee Resources

Apogee Rockets – RockSim Information
https://www.apogeerockets.com/RockSim/RockSim_Information

Apogee Rockets – RockSim Quick Start Guide
https://www.apogeerockets.com/RockSim_Quick_Start_Guide?pg=quickside

Apogee Rockets – RockSim Discounted Temp License
https://www.apogeerockets.com/Rocket_Software/RockSim_Educational_TARC

16e. Tripoli Rocketry Association (TRA) Resources

TRA Website
http://www.tripoli.org/

TRA Membership
http://www.tripoli.org/Membership

TRA Certification Overview
http://www.tripoli.org/Certification

TRA Universal Certification Form
https://tripoli.org/docs.ashx?id=859597

TRA Prefectures
http://www.tripoli.org/Prefectures

16f. National Association of Rocketry (NAR) Resources

NAR Website
https://www.nar.org/

NAR Membership
https://www.nar.org/content.aspx?page_id=22&club_id=114127&module_id=667409

NAR Clubs
https://www.nar.org/content.aspx?page_id=225&club_id=114127

FAA Waiver on NAR Website
https://www.nar.org/content.aspx?page_id=22&club_id=114127&module_id=668326

16g. HPR Launch Information - Canada

https://www.canadianrocketry.org/en/certifications/certifications_L1toL4

17. WSGC Rocket Workshop Evaluation Questionnaire

Thank you for participating in our Launch 2 Learn workshop. Please fill out the workshop evaluation questionnaire here:

https://forms.gle/HVJBr5Jg5sSri7sf8

INTENTIONALLY
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3 Inch Air Frame Fin Guide